Resolution refinement followed by the rodeo algorithm: a gate-level benchmark of eigenstate-preparation routes on the pairing Hamiltonian

Scott Bogner,[1] Cecilie Glittum,[2] Heiko Hergert,[1] Morten Hjorth-Jensen,[2,*] Gunnar Lange,[2] Ryan LaRose,[3] Dean Lee,[1] Francesco Pietro Massel,[4,2] and Jean Paul Sadia[5,6]

[1] *Facility for Rare Isotope Beams and Department of Physics and Astronomy, Michigan State University, East Lansing, MI 48824, USA*
[2] *Department of Physics, University of Oslo, N-0316 Oslo, Norway*
[3] *Department of Computational Mathematics, Science and Engineering and Facility for Rare Isotope Beams, Michigan State University, East Lansing, MI 48824, USA*
[4] *Department of Science and Industry Systems, University of South-Eastern Norway, N-3616 Kongsberg, Norway*
[5] *Department of Computational Mathematics, Science, and Engineering, Michigan State University, East Lansing, MI 48823, USA*
[6] *Center for Quantum Computing, Science, and Engineering, Michigan State University, East Lansing, MI 48823, USA*

(Dated: October 20, 2026)

Background: Projective spectral filters such as the rodeo algorithm prepare eigenstates to exponential accuracy in the number of cycles, at a cost set by the overlap of the input with the target. The routes that can supply that input — variational circuits, an adiabatic ramp, or resolution refinement, the adiabatic transport of an eigenstate from a coarse to a fine single-particle basis — have not been compared on equal terms. **Purpose:** To give a validated, gate-counted reference implementation of resolution refinement followed by the rodeo algorithm on the constant-pairing Hamiltonian, benchmarked at matched accuracy against the same filter fed by the Hartree–Fock determinant, unitary coupled-cluster and ADAPT-VQE states, and an adiabatic ramp. **Methods:** Every stage is compiled to rotations and CNOTs with second-order product formulas and simulated exactly against full configuration interaction. The figure of merit is the ground-state infidelity of the accepted ensemble against the CNOT count per accepted preparation, minimized jointly over evolution time, shift, step count and filter cycles, at eight and at sixteen qubits. **Results:** Refinement raises the overlap of the prolonged determinant from 0.87 to 0.998 with no trained parameter, and the filter removes the remainder geometrically, following the exact Gaussian-ensemble expression derived here. The response norm of the path at its start, $\|Q\,H_{\mathrm{high}}P\psi_{\mathrm{low}}\|^2/b^2$ for shift b, fixes how the shift must scale; the long-run state of the digital filter is the dominant eigenoperator of an averaged post-selection map, which sets its accuracy floor; and the available sufficient adiabatic and Trotter conditions overestimate the cost by two to three orders of magnitude. At the targets studied the cheapest adiabatic operating points use no filter cycle, refinement and the ramp travel at the same speed once the shift is tuned and differ only by the embedded coarse term, and an ADAPT-VQE ansatz grown from the seniority-zero pool reaches 10^{-4} with about a thousand CNOTs at both sizes, two orders of magnitude below every other route. **Conclusions:** Resolution refinement followed by the rodeo algorithm is a validated route to a high-fidelity correlated eigenstate that needs no variational training, but on the pairing model never the cheapest one; the benchmark and its shift rule fix the terms on which its advantages for larger spaces must be demonstrated.

I. INTRODUCTION

Determining the low-lying eigenvalues and eigenvectors of a many-body Hamiltonian is one of the grand challenges of quantum many-body theory, where the Hilbert-space dimension grows exponentially with the number of constituents. Quantum computers offer new routes to eigenvalue estimation and eigenstate preparation, and their prospects for nuclear structure have been mapped in recent community studies [1–3], with schematic nuclear models — pairing, Lipkin, and shell-model Hamiltonians — serving as the standard proving ground [4–10]. The state-of-the-art preparation methods, however, each face an obstacle at scale. The variational quantum eigensolver (VQE) [11] is limited by the cost of its classical optimization loop, by the accuracy floor of its ansatz and, for sufficiently expressive circuits, by barren plateaus — gradients that vanish exponentially with system size [12, 13]. Adiabatic state preparation requires evolution times that diverge if the gap between the tracked state and the rest of the spectrum closes along the path [14]. Projective spectral filters — quantum phase estimation [15], polynomial filters based on quantum signal processing [16, 17], projection by measurement [18], and the rodeo algorithm (RA) [19–23] — converge exponentially in the number of cycles, but only once the initial state has a significant squared overlap p with the target eigenstate; their cost grows as $1/p$, and for a fixed, unoptimized reference state p falls exponentially with system size.

These limitations are complementary, and the idea of *preconditioning* the input of a projective filter by an adi-

* mhjensen@uio.no

abatic stage is as old as the rodeo algorithm itself: the original proposal [19] already combines a short adiabatic ramp with the filter, the hierarchical fusion method of Ref. [24] builds large systems from exactly solvable subsystems by alternating adiabatic ramps and rodeo cycles, and the resolution-refinement proposal [25] explicitly recommends switching from adiabatic evolution to a filter such as the rodeo algorithm once the overlap is large. *Resolution refinement* bootstraps an eigenstate across model-space scales: an eigenstate of a coarse (low-resolution) Hamiltonian is lifted by a prolongation operator into a finer space and adiabatically evolved to the fine-resolution eigenstate. Because refinement changes the low-energy spectrum only mildly, the interpolation path can be kept gapped, and under the additional assumption that the first-order perturbative correction to the tracked state stays bounded [25] the required adiabatic time scales as the inverse gap times a slowly growing system-size factor rather than as the inverse gap squared. The same bootstrapping idea applied along the other axis is *nucleation*, proposed recently by some of us [26]: the exact ground state of a small lattice (the seed) is compiled into a circuit, and the lattice is then grown in stages, each stage a Trotterized adiabatic interpolation that couples newly added qubits to the existing bulk, until the target size is reached. Refinement keeps the domain fixed and raises the resolution; nucleation keeps the resolution fixed and grows the domain. Both replace one hard global preparation by a sequence of local, gapped adiabatic steps that start from an exactly solved problem, both are compiled as Trotterized adiabatic circuits, and Ref. [26] supplies for such circuits sufficient conditions on the evolution time and on the number of Trotter steps, together with a variational variant that reuses the Trotterized circuit as a warm-started ansatz. The present work implements this chain — (i) prepare a low-resolution eigenstate; (ii) prolong and adiabatically refine for a time T; (iii) apply the rodeo algorithm — for a hierarchy of fermionic single-particle bases, and asks the question that decides whether the refinement stage earns its gates: does refinement followed by filtering beat filtering the best input that is available cheaply?

The pairing problem has become a standard target for quantum algorithms, and the program of Lacroix and collaborators is the closest in spirit to ours. Its common thread is the deliberate use of symmetry breaking and restoration: a symmetry-breaking (BCS-like) trial state is cheap and correlated, and the symmetry is restored afterwards by a projective step — quantum phase estimation used as a projector [27], Grover-type oracles [28], spin filters [29] or classical shadows [30] — with the restored state then handed to a spectral post-processing stage such as phase estimation or a Krylov expansion [31, 32]; Refs. [10, 33] review this viewpoint and Ref. [34] the broader many-body landscape. More recently the same group has used ADAPT-VQE on pairing Hamiltonians with symmetry-conserving and symmetry-breaking operator pools [9], extracted excited states from the subspace spanned along the ADAPT convergence path [35], obtained Lipkin spectra with a quantum equation-of-motion approach [5] and with a two-qubit generator-coordinate scheme [36], and built one-body Green's functions of small superfluid systems from a variational even-N ground state and a quantum subspace expansion for its $N\pm1$ neighbors [37]. Our pipeline addresses the step that all of these share — obtaining a high-fidelity eigenstate of the fine-resolution Hamiltonian from a cheap approximate one — with a different mechanism: not symmetry projection or subspace diagonalization, but an adiabatic path across resolutions followed by an energy filter; the relation is taken up in Sec. VI.

In this work we realize the full pipeline end to end on a transparent and exactly solvable testbed, the constant-pairing Hamiltonian familiar from nuclear and condensed-matter many-body theory [38] and integrable by the Richardson ansatz [39–41]. Its exact solvability is precisely what makes it the right laboratory: every stage of the pipeline can be validated quantitatively — the gate-level UCCSD coarse solver against full configuration interaction (FCI) and coupled-cluster doubles (CCD); the prolongation and adiabatic refinement against the exact high-resolution eigenstate; and the rodeo filter against the analytic success-probability formulas of Ref. [20]. All circuits are implemented explicitly as rotation and CNOT gates (and an ancilla-based rodeo cycle), so the demonstration doubles as a reference implementation for hardware.

Relative to the resolution-refinement proposal of Ref. [25], the rodeo literature [19, 20] and the fusion method [24], which established adiabatic preconditioning of the filter on lattice and spin systems, the present work contributes three things. (a) A reference implementation: refinement for a hierarchy of *single-particle bases* in a fermionic pairing problem, where the prolongation is gate-free and the embedded coarse Hamiltonian is a conditioned operator whose circuit we give and count, with every stage — Trotterized refinement and Trotterized controlled evolution included — compiled to rotations and CNOTs, validated against exact results at eight and at sixteen qubits, and reproducible from a single executable notebook. (b) A benchmark on equal terms: the refinement-plus-filter chain is compared with the same filter applied to the unrefined input, to an ordinary adiabatic ramp, to a prolonged coarse variational state and to full-space variational states — the UCCSD template, the restricted pair-rotation ansatz that reproduces it, and an ADAPT-VQE ansatz grown from the seniority-zero pool — with the ground-state infidelity as the common observable, matched evolution accuracy, a two-qubit-gate budget per accepted preparation that separates one-time training, repeated preparation and the filter, and the minimum cost at a given target accuracy, filter cycles included or not. (c) Three results that transfer beyond the model: the response norm of the refinement path at its start, $\|QH_{\rm high}P\psi_{\rm low}\|^2/b^2$ ($P(k-P)g^2/(4b^2)$ for the pairing seed), which fixes how the shift of the path must

scale with the size and the coupling; the characterization of the digital filter's long-run state and acceptance by the dominant eigenoperator of an averaged post-selection map, which sets its accuracy floor and the decay of its acceptance; and the evaluation of the rigorous sufficient conditions on the adiabatic time and on the Trotter step count of Ref. [26] against the resources actually needed. The outcome of (b) is stated plainly. Searched jointly over evolution time, shift and step count, the two adiabatic routes — refinement and an ordinary ramp from the Hartree–Fock Hamiltonian — reach the targets without any filter cycle and travel at the same speed once the shift is tuned, differing only by the cost of the embedded coarse term (a tie at eight qubits, 19% in the ramp's favor at sixteen for $g = 1$ and a tie at $g = 2$); at eight qubits a four-parameter pair-rotation ansatz beats both by two orders of magnitude without any filter, and at sixteen (Sec. V E), where the single-layer ansatz degrades and needs a filter that costs 2–3 times more than the adiabatic routes, an adaptively grown ansatz (Sec. V D) still reaches the same accuracy with a thousand CNOTs. The case for the refinement stage rests on its freedom from classical optimization and training measurements, and on coarse eigenstates that are better than the determinant, which this hierarchy does not exercise.

The pairing force conserves seniority, and that symmetry is what keeps the model transparent: the ground state lives in the seniority-zero subspace, the one-particle–one-hole channel is closed, and single excitations drop out of the variational ansatz. We keep to this seniority-conserving model throughout the present paper, as does most of the quantum-algorithm literature on pairing cited above; a seniority-breaking interaction, which opens the one-particle–one-hole channel of realistic two-body forces and degrades the single-reference variational input, is the natural next test of the benchmark and is discussed in Sec. VI.

II. THE PAIRING MODEL AND ITS QUBIT ENCODING

We study the constant-pairing Hamiltonian

$$H = \delta \sum_{p=0}^{k-1} p \sum_{\sigma=\uparrow\downarrow} n_{p\sigma} - \frac{g}{2} \sum_{p,q=0}^{k-1} a^\dagger_{p\uparrow} a^\dagger_{p\downarrow} a_{q\downarrow} a_{q\uparrow}, \tag{1}$$

with k doubly degenerate single-particle levels of spacing δ (set to $\delta = 1$) and a pairing strength g. The interaction scatters time-reversed pairs between levels; it conserves the seniority (the number of unpaired particles), so the ground state of N paired particles lives entirely in the seniority-zero subspace, and the model is exactly solvable by the Richardson ansatz [39–41] — the property that anchors every benchmark below. We map level p and spin σ to qubit $2p + \sigma$ (so a k-level problem uses $2k$ qubits) and encode the fermionic operators through the Jordan–Wigner transformation. The Hartree–Fock (HF) reference fills the lowest $N/2$ levels, with $E_{\mathrm{HF}}(N{=}4, g) = 2 - g$.

Throughout we use $N = 4$ particles as the running example. The fixed-N sector dimension is $\binom{2k}{N}$, growing from 1 at $k = 2$ (a single filled determinant) to 70 at $k = 4$ and 1820 at $k = 8$; the seniority-zero ground-state problem is correspondingly compact, which makes the model an ideal multi-resolution laboratory. "Basis refinement" here means *increasing k at fixed N*: the number of single-particle levels (and qubits) grows while the particle number is held fixed.

III. THE HYBRID PIPELINE

A. Coarse-state preparation: gate-level UCCSD–VQE

The coarse eigenstate is prepared with a unitary coupled-cluster ansatz restricted to singles and doubles (UCCSD) [42, 43], Trotterized into a product of excitation circuits acting on the HF reference,

$$|\Psi(\boldsymbol{\theta})\rangle = \prod_{\mu} e^{\theta_\mu (\hat{\tau}_\mu - \hat{\tau}^\dagger_\mu)} \, |\mathrm{HF}\rangle \, . \tag{2}$$

Through the Jordan–Wigner map each anti-Hermitian generator becomes a sum of commuting Pauli strings, so every factor compiles into a basis change, a CNOT ladder, a single $R_z(\theta)$ rotation, and the uncompute; a single excitation yields two Pauli-rotation sub-circuits and a double excitation yields eight (the circuits are shown in Fig. 1). The ordered product differs in general from the exponential of the summed cluster operator, so the ansatz is the *factorized* (disentangled) UCCSD, and the operator order — singles first, then doubles, each in the lexicographic order of the pool — is part of its definition. We optimize with the exact analytic gradient of the product ansatz: a fermionic excitation generator has eigenvalues 0 and $\pm i$, so the dependence of the energy on one parameter contains first *and* second harmonics, and

the single-frequency two-term parameter-shift rule is not sufficient in general. Because the pairing force does not couple to seniority-changing single excitations, their energy gradient vanishes identically on the seniority-zero manifold; in the zero-initialized optimizations used here all singles amplitudes stay at zero to machine precision and the solver is effectively a *doubles* theory. (This is a property of the optimization on the symmetry-preserving manifold, not a theorem about every unconstrained optimum of an ordered ansatz; and unitary doubles and classical CCD remain distinct approximations, as the different energies of Table I show.) A variational solver supplies an *approximation* to the coarse ground state, $|\psi_{\rm low}\rangle = c_0\,|E_0^{\rm low}\rangle + \sum_{j>0} c_j\,|E_j^{\rm low}\rangle$ with coarse infidelity $\epsilon_{\rm low} = 1 - |c_0|^2$; the later stages are written for an exact coarse eigenstate, and Sec. III C states what an imperfect input costs.

B. The prolongation operator

The prolongation P embeds a low-resolution state into the high-resolution space. For basis refinement it maps every many-body basis state of the $k_{\rm low}$ space to the *same* occupation pattern in the $k_{\rm high}$ space, leaving the new single-particle levels empty. In canonical Jordan–Wigner ordering this is gate-free; for lattice refinement the analogous operator distributes each occupied coarse site over the neighboring fine sites with two single-qubit rotations and two CNOTs per site per spatial direction, i.e. $O(N_{\rm sites})$ gates per dimension [25]. The prolonged state $|\Phi_0\rangle = P\,|\psi_{\rm low}\rangle$ satisfies $\langle\Phi_0|H_{\rm high}|\Phi_0\rangle = \langle\psi_{\rm low}|H_{\rm low}|\psi_{\rm low}\rangle$ exactly *because* the nested bare pairing Hamiltonians used here obey $P^\dagger H_{\rm high} P = H_{\rm low}$ (same level spacing and coupling at both resolutions); the identity does not follow from the definition of a prolongation and can fail for a renormalized or re-parametrized coarse Hamiltonian. For the exact coarse eigenstate the right-hand side is $E_{\rm low}$, for a variational one its variational energy. That the prolonged state also has a *large overlap* with the high-resolution ground state is a separate, empirical property of this pair of Hamiltonians, documented by the overlaps reported in Sec. IV B; preservation of the coarse expectation value does not imply it.

C. Resolution refinement (adiabatic phase)

The prolonged state is refined by adiabatically following the interpolated, energy-shifted Hamiltonian

$$H(s) = \cos^2\!\Big(\tfrac{\pi s}{2}\Big)\,P(H_{\rm low}-\mu)P^\dagger + \sin^2\!\Big(\tfrac{\pi s}{2}\Big)\,(H_{\rm high}-\mu), \qquad s = \frac{t}{T}, \tag{3}$$

evolved by a Suzuki–Trotter product over total time T. The shift μ (here $\mu = E_{\rm low} + 0.6$) places the tracked low-energy states below the large null space of $P^\dagger$ at $s = 0$; provided the tracked branch stays isolated from the rest of the spectrum along the whole path — a condition that the shift alone does not guarantee and that the spectral calculations of Secs. IV B–IV D verify case by case — the level followed from $s = 0$ to $s = 1$ is the high-resolution ground state. The shift is not a global energy offset: writing $\mu PP^\dagger\cos^2(\pi s/2) = \mu\mathbb{1}\cos^2(\pi s/2) - \mu\cos^2(\pi s/2)(\mathbb{1} - PP^\dagger)$ exhibits it as a relative penalty on the complement of the prolonged subspace, which is why changing μ changes the path dynamics while leaving the end-point spectrum untouched. The first term is *not* the coarse Hamiltonian acting on the coarse register alone: since $PP^\dagger$ projects on the states in which the appended single-particle levels are empty,

$$A \equiv P(H_{\rm low}-\mu)P^\dagger = (H_{\rm low}-\mu)\otimes|0\cdots0\rangle\langle0\cdots0|_{\rm app}\,, \tag{4}$$

a coarse-space operator conditioned on the $2(k_{\rm high}-k_{\rm low})$ new qubits being in $|0\rangle$; its circuit and cost are given in Sec. V B. At $s = 0$ the gap above the tracked state is $\min(E_1^{\rm low} - E_0^{\rm low},\, \mu - E_0^{\rm low})$: increasing the shift lowers the embedded coarse states further below the null space, and the gain in initial gap saturates at the coarse excitation gap (for a one-dimensional coarse space, as in the $k = 2$ start used below, it is simply $\mu - E_{\rm low}$). Along the path the instantaneous spectrum is found to stay smooth and gapped in all cases studied here. A gapped path alone does not fix the adiabatic time scale; Ref. [25] shows, for the affine interpolation $H(u) = (1-u)A + uB$ that Eq. (3) traverses with $u = \sin^2(\pi s/2)$, that when the squared norm of the first-order perturbative correction to the tracked state, $\chi_u = \sum_{n>0} |\langle n|\partial_u H|0\rangle|^2/(E_n - E_0)^2$, stays uniformly bounded along the path, the adiabatic time scales as $T \sim 1/\Delta E$ rather than as the $O(1/\Delta E_{\rm min}^2)$ of the generic adiabatic theorem [14]. (χ_u is a squared eigenvector response, not by itself a leakage probability; the response to the scheduled parameter is $\chi_s = [\frac{\pi}{2}\sin\pi s]^2\chi_u$ and vanishes at both ends.) Section IV D evaluates χ_u for our path family and treats the overlap $p = |\langle\Psi_{\rm high}|\Phi(T)\rangle|^2$ as the figure of merit of the stage. Refinement is unitary and does not purify an imperfect coarse input: if the coarse state carries infidelity $\epsilon_{\rm low}$, ideal adiabatic continuation transports that weight to excited fine states, and the final infidelity of the digitized stage obeys

$$1 - F_{\rm refine} \le \big(\sqrt{\epsilon_{\rm low}} + a_{\rm ad} + \eta\big)^2, \tag{5}$$

where $a_{\rm ad}$ bounds the leakage amplitude of the ideal evolution started from the exact coarse ground state and

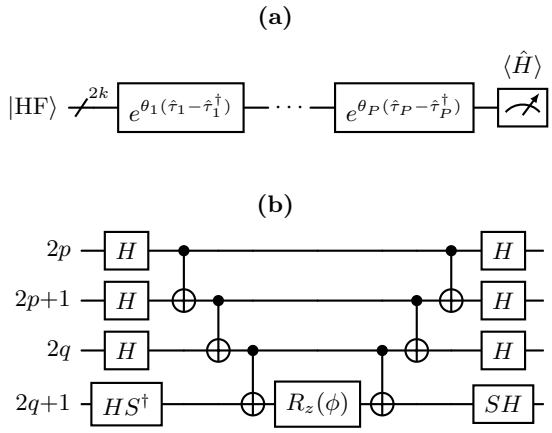


FIG. 1. The variational quantum eigensolver as a circuit. (a) Ansatz level: the Trotterized UCCSD product acts on the HF register ($2k$ qubits, slashed wire) and the energy is estimated by measuring $\langle \hat{H} \rangle$ (in practice a sum of Pauli expectation values, each averaged over many shots); a classical optimizer updates $\boldsymbol{\theta}$ and the entire circuit is re-executed, once per energy or gradient evaluation. (b) Compiled circuit of one weight-4 Pauli rotation, here $e^{-i\frac{\phi}{2}X\otimes X\otimes X\otimes Y}$ acting on the four qubits of a pair-hopping double excitation $(2p, 2p{+}1) \to (2q, 2q{+}1)$: basis changes into the Z eigenbasis (H for X, $HS^\dagger$ for Y), a CNOT parity ladder, the single parametrized $R_z(\phi)$, and the uncompute. Eight such blocks (48 CNOTs) build one pair-hopping double excitation; 160 Pauli rotations of weights 3–8 (1312 CNOTs) build the $k = 4$ preparation of Table I.

η the operator-norm error of the product formula (decompose the input into its ground and orthogonal components and apply the triangle inequality to the final component orthogonal to the target). The three sources must share a target ϵ; assigning all of it to the adiabatic term and then adding the digitization error does not certify the same total. The residual excited weight is what the rodeo stage removes, provided the overlap with the target remains nonzero.

The interpolation (3) is the $\sin^2(\pi s/2)$ schedule, whose vanishing derivatives at both ends remove the boundary terms of the adiabatic bound of Jansen, Ruskai and Seiler [44]; it is also the schedule of the nucleation method [26], where each growth stage couples newly added qubits to a bulk lattice in the way the second term of Eq. (3) couples the appended levels to the coarse space. Ref. [26] turns that bound into explicit sufficient conditions for a Trotterized adiabatic stage: a total time

$$T \;\geq\; \frac{1}{\sqrt{\epsilon}} \int_0^1 \left[\frac{\left\| \ddot{H} \right\|}{\Delta^2} + 7\, \frac{\left\| \dot{H} \right\|^2}{\Delta^3} \right] ds \qquad (6)$$

guarantees a final infidelity at most ϵ, with derivatives taken with respect to s and $\Delta(s)$ the instantaneous gap, and $n_s \geq \eta^{-1}(T\Lambda_1/4 + T^2\Lambda_2/2)$ first-order Trotter steps keep the operator error of the digitized evolution below η, with $\Lambda_1 = \max_s \left\| \dot{H} \right\|$ and, for the many groups of our compilation, $\Lambda_2 = \max_s \sum_{i<j} \|[H_i(s), H_j(s)]\|$, the sum of the commutator norms over all pairs of groups — the natural extension of the two-group statement of Ref. [26]. The time condition applies verbatim to Eq. (3), since the path is twice differentiable; but both conditions are sufficient, not necessary, and Ref. [26] documents for a twelve-site transverse-field Ising chain that they overestimate the time and the step count actually needed by several orders of magnitude — in part because the Trotter error of a digitized adiabatic sweep partly heals itself [45]. The refinement times T^* extracted below are therefore empirical, read off the computed overlap with the exact fine ground state rather than from a bound, and the product-formula step counts of Sec. V B are chosen by the same exact benchmark. The refinement circuit is optimizer-free in execution — no parameter is trained — but its controls (T, μ, n_s) are calibrated here against the exact answer; an implementable calibration would read them from the acceptance statistics of the filter or from a coarse-space estimate of χ, and we have not demonstrated that. Both conditions are evaluated for our path in Sec. S4 of the Supplemental Material [59] and summarized in Sec. IV D.

D. The rodeo algorithm

The rodeo algorithm [19, 20] removes the residual error left by the adiabatic tail. One cycle acts on the system register plus a single ancilla (reused across cycles by mid-circuit measurement; the circuit is shown in Fig. 2): a Hadamard on the ancilla, the controlled evolution e^{-iHt_k} of the system, a phase gate $P(E\,t_k)$ on the ancilla (which effectively shifts $H \to H - E$), a second Hadamard, and an ancilla measurement. Post-selecting the ancilla in $|0\rangle$ applies the non-unitary filter $\frac{1}{2}\left(\mathbb{1} + e^{iEt_k} e^{-iHt_k}\right)$ to the system: an eigenstate $|E_j\rangle$ acquires the factor $e^{i(E-E_j)t_k/2} \cos[(E_j - E)t_k/2]$. For an eigenstate input the probability that all M ancilla mea-

surements return 0 is therefore [20]

$$P_{0^M}\big(E \mid \{t_k\}\big) = \prod_{k=1}^{M} \cos^2\Big[\tfrac{t_k}{2}\big(E_{\rm obj} - E\big)\Big], \qquad (7)$$

and averaging the t_k over a Gaussian of RMS width σ gives

$$P_{0^M}(E) = \left[\frac{1 + e^{-(E_{\rm obj}-E)^2\sigma^2/2}}{2}\right]^M. \qquad (8)$$

On resonance ($E = E_{\rm obj}$) the success probability is unity at every cycle, while off-resonant weight is suppressed geometrically toward the background 2^{-M} (the arithmetic mean of $\cos^2$ is $1/2$). Two readings of that background must be kept apart: the Gaussian-averaged single-cycle factor $q = \frac{1}{2}[1 + e^{-(E_{\rm obj}-E)^2\sigma^2/2}]$ is never smaller than $1/2$, so for the *ensemble* of random schedules $q^M \geq 2^{-M}$ is a lower bound on the retained fraction of any component, reached only far from resonance ($|E_{\rm obj} - E|\sigma \gg 1$); for one *fixed* schedule there is no such bound, since an individual cosine factor can vanish. Scanning E locates eigenvalues as peaks whose width is $\sim 1/\sigma$ at fixed M and, since $q(E)^M \simeq \exp\big[-M\sigma^2(E - E_j)^2/4\big]$ near a peak, $\sim 1/(\sigma\sqrt{M})$ when the cycle number is also varied; fixing E at the target eigenvalue turns the algorithm into a state-preparation filter.

The controlled evolution is the only system-size-dependent ingredient. On hardware e^{-iHt} is realized by a Suzuki–Trotter decomposition over the $2k$ qubits [46–48]; for the single-qubit benchmark of Ref. [20] it reduces to a Z–Y–Z Euler rotation $R_{\hat n}(\theta) = e^{-i\theta\,\hat n\cdot\vec\sigma/2}$ implemented through the Qiskit native gate. In Secs. IV A–IV E we apply the controlled evolution exactly within the conserved N-particle sector to isolate the algorithmic behavior; the circuit-level pipeline with explicit product formulas, and the Trotter error it introduces, are quantified in Sec. V B. Throughout we draw the cycle times t_k from a Gaussian as in Ref. [19]; optimized deterministic [21] or geometric [23] time schedules suppress off-resonant weight faster at fixed runtime and would only improve the numbers reported here, as would the controlled-reversal-gate compilation of the controlled evolution of Ref. [22].

a. Convergence criterion and the random-time ensemble. With E tuned to the target eigenvalue E_0, the all-zero acceptance probability of an input $|\psi\rangle = \sum_k c_k\,|E_k\rangle$ after M cycles, averaged over the Gaussian time sets, is

$$P_M = p + \sum_{k\neq 0} |c_k|^2\, q_k^M, \qquad p = |c_0|^2, \qquad (9)$$

with $q_k = \frac{1}{2}\big(1 + e^{-(E_k-E_0)^2\sigma^2/2}\big)$ the Gaussian-averaged single-cycle factor of Eq. (8). On resonance and with exact evolution the target weight p is preserved at every round while off-resonant components decay geometrically, so $P_M \to p$; at a detuning $d = E - E_0$ the retained target weight is $p\,q_0(E)^M$ and decays with M, so a substantial acceptance requires $|d|\sigma\sqrt{M} \ll 1$ (Sec. IV E tests this), and with a product-formula evolution the preservation is only approximate (Sec. V B). For an exactly degenerate target energy the filter selects the eigenspace, not a vector; and a component with zero initial overlap cannot be created by any function of H. Two fidelities must be distinguished for a randomized filter, whose schedules are drawn afresh for every attempted preparation. For one fixed time set $\{t_k\}$ the post-selected state has a fidelity $\mathcal{F}(\{t_k\})$ and an acceptance probability $P(\{t_k\})$; the fidelity of the *ensemble of accepted runs*, which is what repeated executions on a device deliver, is the acceptance-weighted average $\mathcal{F}_M = \langle P\,\mathcal{F}\rangle/\langle P\rangle$, and for it the spectral decomposition gives exactly

$$1 - \mathcal{F}_M = \frac{\sum_{k\neq 0} |c_k|^2 q_k^M}{p + \sum_{k\neq 0} |c_k|^2 q_k^M} \leq \frac{(1-p)\,q_{\rm max}^M}{p + (1-p)\,q_{\rm max}^M}, \quad (10)$$

with $q_{\rm max} = \frac{1}{2}\big(1 + e^{-\Delta_{\rm min}^2\sigma^2/2}\big)$ and $\Delta_{\rm min}$ the smallest excitation energy carrying weight. The familiar estimate $1 - \mathcal{F}_M \simeq \frac{1-p}{p}2^{-M}$ is the limit $\Delta_{\rm min}\sigma \gg 1$ of Eq. (10); for the pairing spectra used below ($\Delta_{\rm min}\sigma \geq 5$) the two are indistinguishable, and we plot Eq. (10) as the analytic expectation of the ensemble average. All sampled infidelities in this paper are acceptance-weighted; the unweighted mean of $\mathcal{F}(\{t_k\})$ over time sets differs from $\mathcal{F}_M$ by at most $1/p - 1$ relative, i.e. less than one percent for inputs with $p > 0.99$ and a few percent for $p \simeq 0.87$, and the dominant uncertainty of a sampled point is instead the finite number of schedules (Sec. IV E). Reaching an ensemble fidelity $1 - \delta$ needs $M \gtrsim \log_{1/q_{\rm max}}[(1-p)/(p\,\delta)]$ cycles, and since the expected number of repetitions per accepted run is $1/P_M \to 1/p$, the ideal controlled-evolution effort at a resolution ϵ sufficient to isolate the target scales as $O\big(|\log\delta|/(p\,\epsilon)\big)$ [19]. This is not the gate

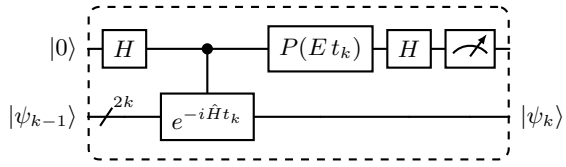


FIG. 2. One cycle of the rodeo algorithm: Hadamard on the ancilla, controlled time evolution $e^{-i\hat H t_k}$ of the $2k$-qubit system register with a Gaussian-random time t_k, ancilla phase $P(Et_k)$, second Hadamard, and a mid-circuit measurement of the ancilla, which is then reset and reused for the next cycle. A run is accepted when all M outcomes read 0; each accepted cycle applies the filter $\frac{1}{2}(\mathbb{1} + e^{iEt_k}e^{-i\hat H t_k})$ to the system. Unlike the VQE of Fig. 1(a), the circuit carries no variational parameters and no classical optimization loop; the controlled evolution, compiled from the same Pauli-rotation blocks as Fig. 1(b) with only the central R_z controlled, is the sole system-size-dependent element ($\approx 4.2 \times 10^2$ CNOTs per Trotter step at $k = 4$).

complexity of the implemented pipeline: input preparation, the search for the target energy, the accuracy of the Hamiltonian simulation and the reduction of the product-formula step with the requested precision are counted separately in Sec. V, and the accuracy floor found there is the direct reason for the distinction. The filter is efficient whenever p is close to one, whatever supplies that overlap — resolution refinement, a variational circuit or a cheap correlated reference — which is why the comparison of Sec. V C is between *inputs* to the same filter.

IV. RESULTS

A. Coarse solver: UCCSD versus exact references

Table I lists the ground-state energies at $g = 1$ for $N = 4$ as the basis grows from $k = 2$ to $k = 8$. The gate-level UCCSD circuit reproduces the operator-level (statevector) UCCSD to machine precision and stays variational ($E_{\mathrm{UCCSD}} \geq E_{\mathrm{FCI}}$), tracking FCI to $\sim 10^{-3}$–10^{-2}, whereas CCD over-binds. Figures S1 and S2 of the Supplemental Material [59] compare all solvers (exact, CCD, statevector and gate-level UCCSD, and a hardware-efficient ansatz [49]) as a function of k: the hardware-efficient ansatz is exact at $k = 2$, reaches FCI at $k = 3$ only after a multi-start optimization and fails at $k = 4$ — the symmetry-blind landscape that motivates the physically structured UCCSD circuit — while the UCCSD solver remains a tight variational bound all the way to $k = 8$ (sixteen qubits), at the cost of a CNOT count that grows steeply (1312 CNOTs already at $k = 4$; Sec. V). Two remarks keep this table in proportion. First, an energy error is not an infidelity: for a normalized state $\langle H\rangle - E_0 = \sum_{j>0} |c_j|^2 (E_j - E_0) \geq \Delta\,(1 - \mathcal{F})$, so with the physical gap $\Delta = 1.823$ the $k = 4$ energy error 1.44×10^{-3} already caps the infidelity at 7.9×10^{-4}; the actual value, obtained from the optimized state, is $1 - \mathcal{F}_{\mathrm{UCCSD}} = 1.78 \times 10^{-4}$, a factor eight below the energy error, and it is this infidelity that every comparison below uses. Second, the scan in k at fixed $N = 4$ is a basis-size study — the fixed-N sector grows as k^4 and the seniority-zero sector as k^2 — and it concerns the UCCSD solver alone, not the complete pipeline; it is not evidence of favorable many-particle scaling at fixed filling.

TABLE I. Ground-state energies at $g = 1$, $N = 4$, for the constant-pairing Hamiltonian. FCI is the exact N-sector result; CCD is coupled-cluster doubles; UCCSD is the gate-level circuit of Sec. III A. "#CNOT" is the two-qubit gate count of the UCCSD state preparation.

k	qubits	FCI	CCD	UCCSD	#CNOT
2	4	1.000000	1.000000	1.000000	0
3	6	0.794697	0.794697	0.794697	272
4	8	0.635548	0.630443	0.636987	1312
6	12	0.372664	0.350058	0.378719	–
8	16	0.145559	0.097263	0.157856	–

B. Resolution refinement: overlap and energy

At $N = 4$ the low-resolution ($k = 2$) space is a single filled determinant, so the prolonged state is the high-resolution HF reference, with first-guess overlaps 0.917 for refinement to $k = 3$ and 0.868 to $k = 4$. Adiabatic evolution along the gapped path of Eq. (3) drives both quantities toward their exact targets: the overlap rises to $\simeq 0.999$ and the energy falls from $E_{\mathrm{HF}} = 1$ to the FCI value from above — 0.7964 for $k = 3$ (FCI 0.794697) and 0.6397 for $k = 4$ (FCI 0.635548) at $T = 30$ (Fig. S3 of the Supplemental Material [59]). Energies above FCI are guaranteed by the variational principle for any normalized state; the improvement with T is what is computed, and monotonicity at every time is not implied. The "exact-slice" propagation still approximates the continuously changing Hamiltonian by 160 midpoint slices; refining to 1 280 slices changes the overlap by 1.5×10^{-6} at $T = 25$ ($0.9975713 \to 0.9975698$) and by 1.1×10^{-6} at $T = 30$, far below the adiabatic infidelities of order 10^{-3} discussed below. The instantaneous spectrum stays open along the whole path (Fig. 3, right), which makes adiabatic reasoning applicable but does not by itself select the $T \sim 1/\Delta E$ cost law discussed in Sec. III C; the larger $k = 4$ space starts lower in overlap and converges more slowly.

C. A nontrivial coarse stage: the $k = 3 \to 4$ chain

At $N = 4$ the $k = 2$ sector is a single determinant, so in the chain above the coarse "solver" is gate-free and the pipeline's first stage does no work. To exercise the pipeline with a working coarse stage we start instead from the gate-level UCCSD state at $k = 3$ — a genuine 272-CNOT variational preparation, indistinguishable from FCI at this resolution (Table I) — prolong $3 \to 4$, and refine along the same path, with the shift μ now referenced to the *variational* coarse energy $\langle\psi_{\mathrm{low}}|H_{\mathrm{low}}|\psi_{\mathrm{low}}\rangle$, the only quantity a device can know. Figure S4 of the Supplemental Material [59] compares the two chains. The better coarse start raises the prolonged overlap from 0.868 to 0.964 and reaches the target overlap 0.99 at $T^* = 8$ (overlap 0.9907; the grid of Fig. 4 gives 9) instead of $T^* = 16$. The $k = 3$ coarse state is essentially exact ($1 - F < 10^{-9}$), so this chain exercises a working coarse stage but does not test the pipeline against a non-negligible coarse-solver error. Whether the shorter sweep saves gates is decided by the embedded coarse term of Eq. (4), which is not free when the coarse space is non-trivial. For the $2 \to 4$ chain A is a single four-qubit conditional phase (14 CNOTs per application, Sec. V B); for the $3 \to 4$ chain it is the complete $k = 3$ Hamiltonian conditioned on the two appended qubits being empty,

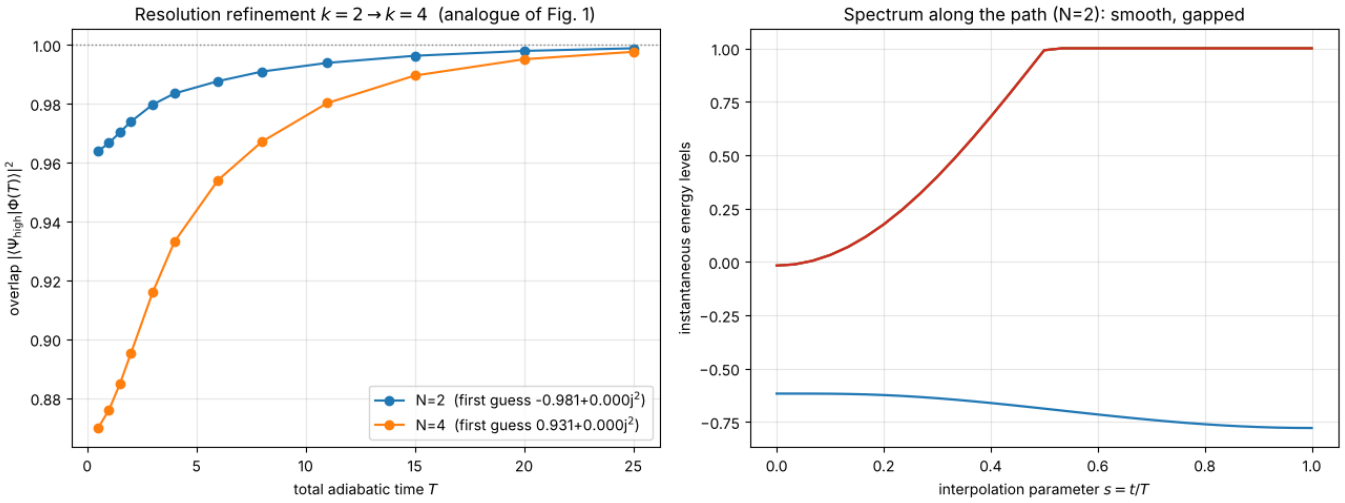


FIG. 3. Resolution refinement $k = 2 \to k = 4$. Left: overlap of the evolved state with the exact high-resolution ground state versus the total adiabatic time T, for $N = 2$ and $N = 4$. Right: the instantaneous spectrum along the interpolation parameter $s = t/T$ stays smooth and gapped, the condition for efficient refinement.

about 350 CNOTs per application, more than the 296 CNOTs of the high-resolution pairing step itself. Because $H_{\rm low}$ contains non-commuting terms, the counted circuit does not exponentiate A as one block but splits it into its own coarse groups (the conditioned diagonal term and three conditioned pair-hopping terms), and we validate the chain at that level in Sec. V B: 22 second-order steps at $T = 8$ give $p = 0.9912$, so the $3 \to 4$ chain costs $2 \times 22 \times (296 + 350) + 96 \simeq 2.9 \times 10^4$ CNOTs, the 96 being the coarse preparation (two pair rotations prepare the three-dimensional seniority-zero coarse space exactly; the 272-CNOT UCCSD template prepares the same state). At the step density of Table II the $2 \to 4$ chain needs 43 steps for $T^* = 16$, i.e. $2 \times 43 \times (296 + 14) \simeq 2.7 \times 10^4$ CNOTs. The shorter adiabatic time of the better start is thus offset by the cost of carrying the coarse Hamiltonian along the path, and the two chains cost the same within 10%; neither figure is an optimized minimum. Starting higher in the hierarchy remains the right choice when the prolonged overlap of the trivial start is poor (the strong-coupling regime of Sec. IV D), but it is not a gate saving in itself.

D. Coupling scan and the path gap

All results above sit at $g = 1$; Fig. 4 probes the refinement across couplings and tests the $T \sim 1/\Delta E$ claim directly. Scanning $g = 0.3$–4 at fixed $\mu = E_{\rm low} + 0.6$ (left panel), the minimum instantaneous gap along the path stays open and nearly constant — it is set by the shift itself, $\Delta E_{\rm min} \simeq \mu - E_{\rm low}$, dropping only to 0.57 at $g = 4$ — while the time to reach overlap 0.99 grows from $T^* = 1$ at $g = 0.3$ to $T^* = 65$ at $g = 4$. The growth goes together with the loss of initial overlap of the prolonged state, which falls from 0.99 to 0.38 over the same range, consistent with a larger change of the eigenstate along the path; but changing g also changes the transition matrix elements and the eigenvector path, so the scan does not isolate the overlap as the cause (the shift scan below changes T^* at fixed initial overlap). It does establish that the nearly constant minimum gap does not determine the required time by itself, which identifies the practical limiter at strong coupling as the starting state rather than a closing gap; whether a coarse start higher in the hierarchy (Sec. IV C) is then the better route is a question of total cost, including the coarse preparation and its conditioned Hamiltonian, not of sweep time alone.

Because the path gap is controlled by the shift, varying μ at fixed $g = 1$ changes the minimum gap over a factor five (right panel). The four finite threshold times, $(\Delta E_{\rm min}, T^*) = (0.3, 50)$, $(0.6, 16)$, $(1.0, 8)$, $(1.5, 6)$, are fitted with an intercept by $T^* \approx 17.1/\Delta E_{\rm min} - 8.4$ (sum of squared residuals 28) and, equally legitimately, by $T^* \approx 4.1/\Delta E_{\rm min}^2 + 4.2$ (residuals 0.2); four threshold times read off a coarse time grid, along a family of paths whose operator changes with μ, do not identify an exponent, and in particular do not exclude inverse-square behavior. We therefore present the scan as a sensitivity study. The response norm χ of Sec. III C makes the limitation explicit (Fig. S5 of the Supplemental Material [59]). With $u = \sin^2(\pi s/2)$ the path is $H(u) = (1 - u)A + uB$ with $A = -b\,|\mathrm{HF}\rangle\langle\mathrm{HF}|$, $b = \mu - E_{\rm low}$, and at $u = 0$ the HF determinant couples through $\partial_u H$ to four pair-transfer configurations with amplitude $-g/2$, all at excitation energy b, so $\chi_u(0) = g^2/b^2$ exactly: 11.1 at $b = 0.3$ and 0.44 at $b = 1.5$, a factor 25, and the maximum of $\chi_u(s)$ sits at $s = 0$ for every shift (the scheduled response χ_s peaks at $s = 0.13$–0.35 and varies by a factor 3.4, from 1.40 to 0.41). The quantity that the inverse-gap argument of Ref. [25] assumes uniformly bounded is thus it-

self varied by the shift scan, which cannot then test that argument by attributing the timing changes to the gap alone. Two further gaps should be kept apart from the artificial shift-controlled gap at $s = 0$: the physical gap of $H_{\rm high}$ in the $N = 4$ sector (1.823 at $g = 1$), a conservative all-sector bound, and the dynamically accessible seniority-zero excitation gap at $s = 1$ (2.300), which is the one the transition matrix elements see. The same scan doubles as the sensitivity study for the shift: the choice $\mu_{\rm buf} = 0.6$ used throughout is conservative, and enlarging the buffer to 1.5 *shortens* the refinement to $T^* = 6$. For a one-dimensional coarse space there is no upper limit to the buffer other than the growth of the energy scale (and hence of the Trotter error) of the path; for coarse spaces with several states the useful initial gap saturates at the coarse excitation gap, as noted in Sec. III C. Section V C adds the limiting case of this observation: an ordinary adiabatic ramp from the diagonal (HF) Hamiltonian, which has no shift and whose seniority-zero gap at $s = 0$ is 2.0, refines the same determinant in a quarter of the time.

a. Sufficient conditions versus practice. The conditions of Sec. III C can be evaluated exactly for the path of Eq. (3), since $\dot{H}$ and $\ddot{H}$ are both proportional to $B - A$ with $A = P(H_{\rm low} - \mu)P^\dagger$ and $B = H_{\rm high} - \mu$; Sec. S4 of the Supplemental Material [59] gives the integrand, the gap along the path and the numbers. With $\|B - A\| = 7.62$ and an integral of 511 in Eq. (6), the time condition demands $T \geq 5.1 \times 10^3$ for an infidelity of 10^{-2} where the computed overlap reaches 0.99 at $T^* = 16$, an overshoot of 3×10^2; the first-order step condition demands 3.4×10^4 steps along the T^* path where 160 first-order or 40 second-order steps reproduce the exact-slice overlap, a factor 2×10^2. The two step counts answer different questions — the bound certifies an operator-norm error of the digitized unitary, one reproduced overlap does not — and the overshoot has an identifiable source: the bound pays for the full norm $\|B - A\|$ and the global gap, whereas the adiabatic error is governed by the individual derivative couplings of the tracked state, which at $s = 0.25$ are 0.60, 0.37, 0.00, 0.11 and 0.08 to the five accessible excited states. These are the fermionic counterpart of the ratios Ref. [26] reports for nucleation of an Ising chain (6×10^1 and 10^7): the conditions certify convergence but do not predict cost, and the refinement times and step counts used below are therefore read off the computed overlap with the exact ground state.

E. Rodeo filtering of the refined state

We verified the rodeo primitive in two ways: the explicit ancilla circuit agrees with the compact filter $\frac{1}{2}(\mathbb{1} + e^{iEt}e^{-iHt})$ to 10^{-16}, and the all-zero probabilities reproduce Eqs. (7)–(8), with the Gaussian-averaged single-cycle factor matching $\frac{1}{2}(1 + e^{-(\Delta E\,\sigma)^2/2})$ to three digits.

The state fed to the filter is the resolution-refined $k = 4$ state of Sec. IV B at $T = 25$ with exact slices and exact controlled evolution, with overlap $p = 0.9976$ with the exact ground state; for contrast we also use the unrefined prolonged guess Φ_0 ($p = 0.868$). This is an *ideal* operating point, chosen to exhibit the filter; the digital pipeline of Sec. V uses other refinement times and a product-formula evolution, and Table S2 of the Supplemental Material [59] keeps the two apart. Scanning the target energy E ($M = 4$ cycles, 12 time sets per energy; Fig. S7 and Sec. S5 of the Supplemental Material [59]) produces a peak at the ground-state energy: a coarse scan ($\sigma = 2$) resolves it from the excited band and a fine scan ($\sigma = 8$, centered on the coarse-scan maximum, i.e. chosen without knowledge of E_0) locates it at $E = 0.6362$ against the exact 0.635548; with 10^3–10^4 shots per energy the peak estimator returns 0.637 ± 0.001, its systematic error exceeding the shot noise. The scan is part of the cost of the method — about 2×10^3 four-cycle runs for the fine scan alone, each repeated over enough shots to estimate an acceptance probability.

Fixing $E = E_0$ and applying M cycles with $\sigma = 3$ (Fig. 5; 40 Gaussian time sets, the same sets for both inputs, generator seed 7 reset per input), the fidelity rises geometrically and the cumulative acceptance settles at $\approx p$, in agreement with Eqs. (9)–(10). For the refined input the exact Gaussian-ensemble infidelity of Eq. (10) is 1.2×10^{-3} after one cycle, 6.1×10^{-4} after two, 3.8×10^{-5} after six and 2.4×10^{-6} after ten, and 12 cycles are needed to reach 10^{-6}; the 40-schedule sample gives 1.35×10^{-3}, 6.5×10^{-4}, 2.1×10^{-5} and 2.0×10^{-6}. The sample lies 45% below the ensemble value at six cycles, which is ordinary finite sampling of products of random attenuation factors: the relative standard error of a 40-schedule mean of the residual weight, computed from 2×10^4 schedules, is 13% at $M = 2$, 37% at $M = 6$ and 82% at $M = 10$ (the error bars of Fig. 5), so the sampled points are statistically consistent with the curve but carry no precision at the smallest infidelities, and the quoted infidelities of this paper are the ensemble values unless stated otherwise (the acceptance weighting itself changes the sampled infidelity by at most $1/p - 1 = 0.24\%$ at this p). The prolonged input reaches 1.3×10^{-4} after ten cycles (ensemble value 1.5×10^{-4}) while its acceptance holds at 0.868. Each cycle buys roughly a factor two in infidelity, so reaching 10^{-6} from $p = 0.998$ takes of order ten cycles, not one or two. Two practical checks complete the picture. First, the target energy on a device is the scan estimate, not the exact eigenvalue: repeating the filter at the scan estimate ($E - E_0 = 7 \times 10^{-4}$, $|d|\sigma\sqrt{M} = 0.005$ at six cycles) changes no digit of the ensemble infidelities quoted above. Second, a deliberately detuned target attenuates the ground state as well: for $E - E_0 = 0.05$, 0.1 and 0.2 the six-cycle ensemble infidelity is unchanged within 3%, 14% and a factor 1.7, while the acceptance falls from 0.998 to 0.965, 0.873 and 0.596 (exact ensemble values) — in this regime the acceptance, not the fidelity, is the sensitive diagnostic of a wrong target, and it then no longer tends to the input overlap; a detuning toward

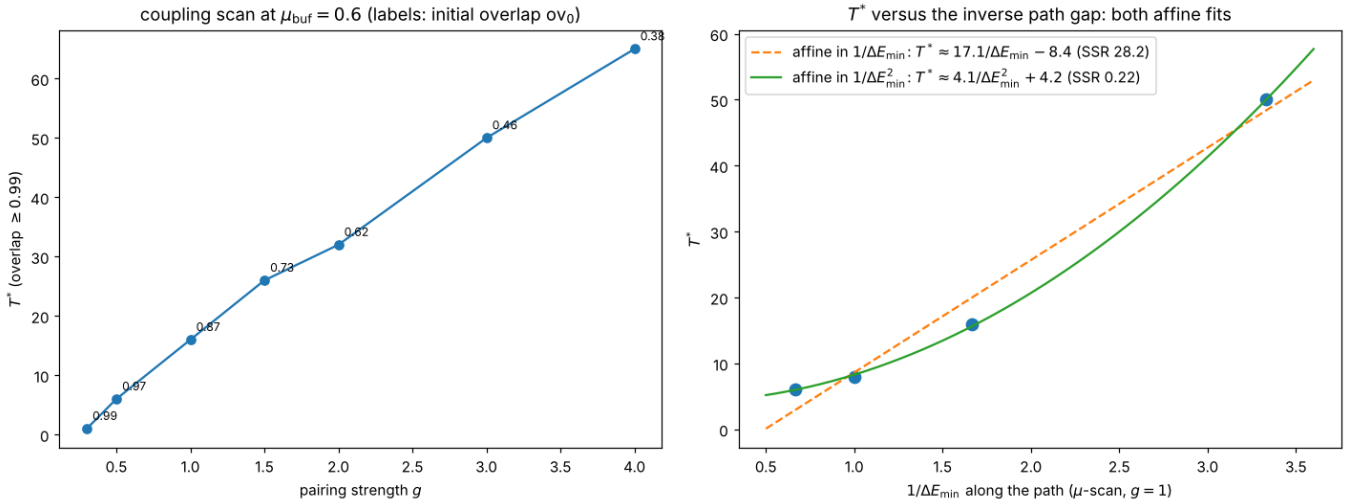


FIG. 4. Left: adiabatic time T^* to reach overlap 0.99 for the $k = 2 \to 4$ refinement versus pairing strength g (point labels: initial overlap of the prolonged state). The minimum path gap stays $\simeq 0.6$ throughout while the starting overlap degrades. Right: T^* versus the inverse minimum path gap, varied through the shift buffer $\mu_{buf} = 0.3$–1.5 at $g = 1$, with affine fits in $1/\Delta E_{min}$ (dashed) and in $1/\Delta E^2_{min}$ (solid); four points on a coarse time grid fit either form and identify no exponent.

another eigenvalue would instead select that state, which the acceptance alone does not certify against.

V. QUANTUM CIRCUITS AND DEPTH COMPARISON

All quantum operations in this work are compiled to a common gate set of single-qubit rotations and CNOTs, so the resource costs of the different routes to the high-resolution ground state — a full-space UCCSD–VQE, the refinement-plus-filter pipeline, and the filter applied directly to cheaper inputs — can be compared gate by gate. Every number quoted below is produced in the accompanying notebook. Two levels of simulation must be distinguished. The Pauli-rotation primitive, the UCCSD excitation circuits, the hardware-efficient ansatz and the ancilla-based rodeo cycle are simulated *gate by gate* on the full $2k$-qubit register and verified against exact exponentials. The product-formula evolutions of Sec. V B, by contrast, are simulated *block by block*: each term of the Hamiltonian splitting is a set of mutually commuting Pauli strings whose exponential is realized exactly by the corresponding rotation–CNOT block, and we apply that exact block exponential within the conserved N-particle sector. This reproduces faithfully the state that the compiled circuit would produce, and the CNOT counts are those of the compiled blocks, but it does not execute every individual rotation and CNOT; we therefore quote CNOT *counts* rather than scheduled circuit depths, which depend on the connectivity and parallelism of a device.

A. Circuit building blocks

The elementary primitive is the Pauli rotation $e^{-i\frac{\phi}{2}P}$ for a Pauli string $P = \bigotimes_q \sigma_q$ [50]: each active qubit is rotated into the Z eigenbasis (H for X, $HS^\dagger$ for Y), a CNOT ladder accumulates the parity onto the last active qubit, a single $R_z(\phi)$ is applied, and the ladder and basis changes are uncomputed. A string of weight w therefore costs $2(w-1)$ CNOTs and one parametrized rotation; the circuit reproduces the exact exponential to 3×10^{-16}.

a. UCCSD preparation circuit. Through the Jordan–Wigner map a single excitation compiles into 2 Pauli rotations whose weight grows with the distance between the spin orbitals (8 CNOTs for adjacent levels, 16 for the $0 \to 4$ single at $k = 3$), and a generic double excitation into 8 rotations (48–64 CNOTs). The doubles that dominate the pairing problem are special: for a time-reversed pair moving between levels p and q the operators act on the adjacent qubits $(2p, 2p{+}1)$ and $(2q, 2q{+}1)$, so the internal Jordan–Wigner Z strings cancel and all 8 Pauli strings have weight 4 — 48 CNOTs per pair-hopping term *independent of the level separation*. The full state preparation at $N = 4$, $g = 1$ then costs 8 excitation operators, 40 rotations and 272 CNOTs at $k = 3$, and 26 operators, 160 rotations and 1312 CNOTs at $k = 4$ (Table I); at $k = 2$ the ansatz is empty (the sector is a single determinant) and the preparation is gate-free.

Figure 1 shows both levels of this hierarchy: the VQE at the ansatz level, whose classical feedback loop wrapped around the full-depth preparation multiplies the 1312-CNOT cost into the total gate budget of the variational route, and the compiled interior of one weight-4 Pauli rotation, the sub-block that occurs 160 times in the $k = 4$

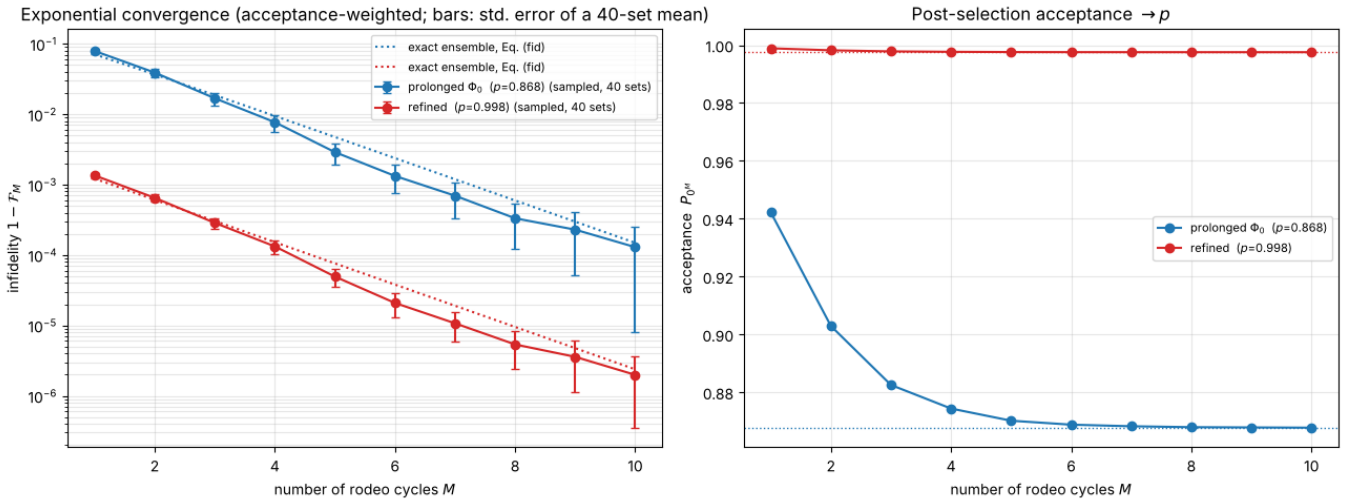


FIG. 5. Rodeo eigenstate preparation at $E = E_0$ ($k = 4$, $N = 4$, $g = 1$, $\sigma = 3$), averaged over 40 Gaussian time sets, for the resolution-refined input ($T = 25$, exact slices, $p = 0.9976$) and the prolonged determinant ($p = 0.868$), with exact controlled evolution and the same 40 time sets for both inputs. Left: acceptance-weighted infidelity of the accepted runs versus the number of cycles M, with error bars giving the standard error of a 40-schedule mean (from 2×10^4 schedules); dotted: the exact Gaussian-ensemble expectation of Eq. (10). Right: the post-selection acceptance converges to the input overlap p.

preparation.

b. Rodeo cycle. One rodeo cycle (Fig. 2) adds a Hadamard on the ancilla, the controlled evolution e^{-iHt_k}, the ancilla phase $P(Et_k)$ — the element that effectively shifts $H \to H - E$ — a second Hadamard, and a mid-circuit measurement; the single ancilla is reset and reused across the M cycles. The rodeo circuit contains no variational parameters and closes no optimization loop: the times t_k are drawn once from a Gaussian, the target energy E is a classical setting, and a run is accepted when all M ancilla outcomes read 0, which applies the filter of Sec. III D cycle by cycle. The controlled evolution is compiled from the same Pauli-rotation primitive: in each rotation sub-circuit only the central R_z needs to be controlled (if the rotation is absent the basis changes and CNOT ladder cancel), and a controlled R_z costs 2 extra CNOTs. For the pairing Hamiltonian (1) a first-order Trotter step contains the $k(k-1)/2$ pair-hopping terms (8 weight-4 rotations, 48 CNOTs each), k diagonal ZZ rotations (2 CNOTs each) and $2k$ single-qubit phases, i.e.

$$N_{\mathrm{CNOT}}^{\mathrm{step}}(k) = 24\,k(k-1) + 2k \qquad (11)$$

CNOTs uncontrolled — 296 at $k = 4$ — and $\approx 4.2 \times 10^2$ CNOTs per step once the controls are added. A rodeo cycle with n_T Trotter steps therefore costs $\approx 4.2 \times 10^2\, n_T$ CNOTs at $k = 4$.

B. The Trotterized pipeline

The results of Sec. IV E used exact sector exponentials; on hardware both the refinement slices and the controlled evolution are product formulas, and their error must be quantified at the circuit level. We split the Hamiltonian exactly as the gate compilation does: a diagonal term D (kinetic energy plus the $p{=}q$ pairing term; single-qubit phases and ZZ rotations) and one pair-hopping term h_{pq} per level pair. The eight Pauli strings of each h_{pq} commute, so every factor $e^{-ih_{pq}\delta t}$ is realized *exactly* by its 48-CNOT block, and the only Trotter error is the non-commutativity between terms — exactly as on a device. At $k = 4$, $t = 1$ the second-order (symmetric) formula reaches spectral-norm error 3.4×10^{-2} with 4 steps and 8.3×10^{-3} with 8.

a. The embedded coarse term. The refinement path, Eq. (3), contains a third kind of factor, $e^{-i\cos^2(\pi s/2)A\,\delta t}$ with $A = P(H_{\mathrm{low}} - \mu)P^\dagger$ of Eq. (4). Gate-free prolongation does not make this evolution gate-free. For the $2 \to 4$ chain at $N = 4$ the coarse space is the single determinant $|\mathrm{HF}\rangle$, so $A = (E_{\mathrm{low}} - \mu)\,|\mathrm{HF}\rangle\langle\mathrm{HF}| = -0.6\,|\mathrm{HF}\rangle\langle\mathrm{HF}|$, and within the $N = 4$ sector $|\mathrm{HF}\rangle$ is the unique state with the four appended qubits 4–7 all in $|0\rangle$. Its exponential is therefore a phase conditioned on four qubits being empty, $X^{\otimes 4}\,C^3P(\phi)\,X^{\otimes 4}$ with $\phi = 0.6\cos^2(\pi s/2)\delta t$. Expanding the projector, $|1111\rangle\langle 1111| = 2^{-4}\sum_S (-1)^{|S|} Z_S$ over the 15 nonempty subsets S of the four qubits, gives a phase polynomial of 15 commuting Z-strings; ordered along a Gray code it compiles into 14 CNOTs and 15 R_z rotations (the naive ladder-per-string construction costs 34 CNOTs). We verified that this operator reproduces the sector matrix exponential of A exactly. For a chain that starts from a nontrivial coarse space the term is $(H_{\mathrm{low}} - \mu)$ on the coarse register conditioned on the appended qubits being empty; for the $3 \to 4$ chain each of the 33 Pauli rotations of a $k = 3$ step (single-qubit phases included, since they are no longer free) acquires a double control (6 CNOTs per rotation with a three-qubit

Gray-code phase polynomial), and the constant part of $H_{\mathrm{low}} - \mu$ becomes a conditioned phase (2 CNOTs) rather than a global one, for 350 CNOTs per application, the figure used in Sec. IV C. These are the costs of an ancilla-free reference compilation, not lower bounds: with $c-1$ clean work qubits the conjunction of the c appended controls can be computed once into a flag, every rotation singly controlled by it and the flag uncomputed, which gives 224 CNOTs per application of the $3 \to 4$ term, 1 028 instead of 4 886 for the $6 \to 8$ term of Sec. V E and 44 instead of 254 for its filled $4 \to 8$ term, at the price of a register wider than the $2k+1$ qubits assumed throughout. All refinement budgets below include the coarse term in the ancilla-free convention; for the $2 \to 4$ chain it adds 14 CNOTs to the 296 of a first-order step, i.e. $2 \times 80 \times 14 = 2\,240$ CNOTs or 5% over the $T = 30$ path.

Figure 6 shows the two pipeline stages under this compilation. For the refinement (left), $n_s = 80$ second-order steps along the $T = 30$ path reproduce the exact-slice result to within its own adiabatic error, delivering $p = 0.9977$ against the exact-slice 0.9988; first-order steps need several hundred. For the rodeo stage (right) the honest scheme fixes the maximum Trotter step *size* δt, so a cycle with Gaussian time t_k uses $n_k = \lceil |t_k|/\delta t \rceil$ steps of size t_k/n_k ($\sigma\sqrt{2/\pi}/\delta t \approx 13$ steps on average at $\sigma = 4$, $\delta t = 0.25$). The filter is strikingly diagnostic of its own Trotter error: at $\delta t = 1$ the corrupted evolution is largely *rejected* — the acceptance collapses to 0.10 after six cycles and the fidelity does not approach the target accuracy in the displayed range — while at $\delta t = 0.25$ the infidelity tracks the exact-evolution curve down to 1.2×10^{-4} at $M = 6$ with acceptance 0.989. The intermediate $\delta t = 0.5$ settles at a floor of 2×10^{-3}. The precise statement behind these floors is the following. With exact evolution on resonance every post-selection operator $K_t = \frac{1}{2}(\mathbb{1} + e^{iE_0 t} e^{-iHt})$ leaves the ground-state amplitude unchanged and all K_t share the eigenvectors of H, so the accepted ensemble converges to the ground state and the acceptance to p. With the product-formula evolution $\tilde{U}(t)$ the operators $\tilde{K}_t = \frac{1}{2}(\mathbb{1} + e^{iE_0 t}\tilde{U}(t))$ no longer preserve the true ground state, and since the schedules are drawn afresh for every attempt the accepted ensemble after M cycles is $\rho_M = \mathcal{K}^M(\rho_{\mathrm{in}})/\operatorname{Tr}\mathcal{K}^M(\rho_{\mathrm{in}})$ with the averaged map $\mathcal{K}(\rho) = \mathbb{E}_t[\tilde{K}_t \rho \tilde{K}_t^\dagger]$. The long-run state is the dominant eigenoperator of this trace-decreasing map — not the ground state of one effective Hamiltonian, and not a mixture of the separate filters' ground states — and its eigenvalue is the long-run acceptance per cycle (assuming, as the power iteration verifies here, that the dominant eigenvalue is nondegenerate and its eigenoperator is reached from the input). The expectation over the cycle times must respect the integrand: the product formula changes at every boundary $|t| = n\,\delta t$, so we integrate the half-Gaussian piecewise between these boundaries with a Gauss–Legendre rule whose refinement changes no quoted digit, and report the residual of the eigen-equation ($< 10^{-12}$); a global Gauss–Hermite rule with tens of nodes is not converged for this integrand and, at sixteen qubits, produces spurious dominant states. Evaluated in the six-dimensional seniority-zero block (notebook, Secs. 19 and 22) the map gives an asymptotic infidelity of 1.1×10^{-4} at $\delta t = 0.25$ with eigenvalue 0.9978, 1.9×10^{-3} at $\delta t = 0.5$ with eigenvalue 0.965, and 0.15 at $\delta t = 1$ with eigenvalue 0.63; these reproduce the floors of Fig. 6 and show that the long-run acceptance of the digital filter decays geometrically, by 0.2% per cycle at $\delta t = 0.25$, rather than tending to p as the ideal theorem states. A floor is an asymptote: a finite-cycle track of an input that starts below it is read off the track, not inferred. Coherent product-formula errors thus both shift the located energy and bias the prepared state; the rejection of grossly inaccurate evolutions seen at $\delta t = 1$ is a useful diagnostic, not a guarantee that smaller coherent errors are removed.

C. Gate budget at matched fidelity: which input should the filter get?

The question that decides the value of the refinement stage is not whether the pipeline beats an *unfiltered* variational state — a few rodeo cycles improve any input with large overlap — but whether refinement followed by the filter beats the same filter applied to the best input that is available cheaply, or no filter at all. We therefore feed one and the same Trotterized filter (second order, $\delta t = 0.25$, $\sigma = 4$, $|t_k|$ drawn from a Gaussian, 40 common time sets drawn afresh per attempted preparation) eight inputs: (i) the prolonged $k = 2$ determinant, $p = 0.868$, gate-free; (ii) the four-pair restricted ansatz of Sec. V D, $p = 0.99982$, 192 CNOTs; (iii) the full-template gate-level UCCSD state, the same state for 1 312 CNOTs; (iv) the prolonged $k = 3$ coarse state of Sec. IV C, $p = 0.964$, prepared with two pair rotations (96 CNOTs) and fed to the filter *without* refinement; (v) an ordinary adiabatic ramp from the diagonal Hamiltonian, $H(s) = D + \sin^2(\pi s/2)\sum_{pq} h_{pq}$ started from the HF determinant, which has no shift and no coarse term and whose seniority-zero gap at $s = 0$ is 2.0: it reaches $p = 0.9987$ at $T = 4$ with exact slices and $p = 0.9978$ with 11 second-order steps, $2 \times 11 \times 296 = 6\,512$ CNOTs; (vi) the Trotterized refined state at $T^* = 16$ (43 second-order steps, $p = 0.9922$, $2 \times 43 \times 310 = 26\,660$ CNOTs including the coarse term); (vii) the refined state at $T = 30$ of Fig. 6 (80 steps, $p = 0.9977$, 49 600 CNOTs); and (viii) the $3 \to 4$ chain of Sec. IV C at the counted circuit level ($T = 8$, 22 steps with the coarse term split into its coarse groups, $p = 0.9912$, 28 520 CNOTs). Inputs (iv) and (v) are the two baselines needed to isolate what the specific refinement path contributes: (iv) asks whether refinement is needed after a useful coarse solution, (v) whether the shift-controlled projector path does anything that generic adiabatic preconditioning from the same determinant does not. For each input and each cycle number $M = 0, \ldots, 6$ we track the ensemble infidelity of the

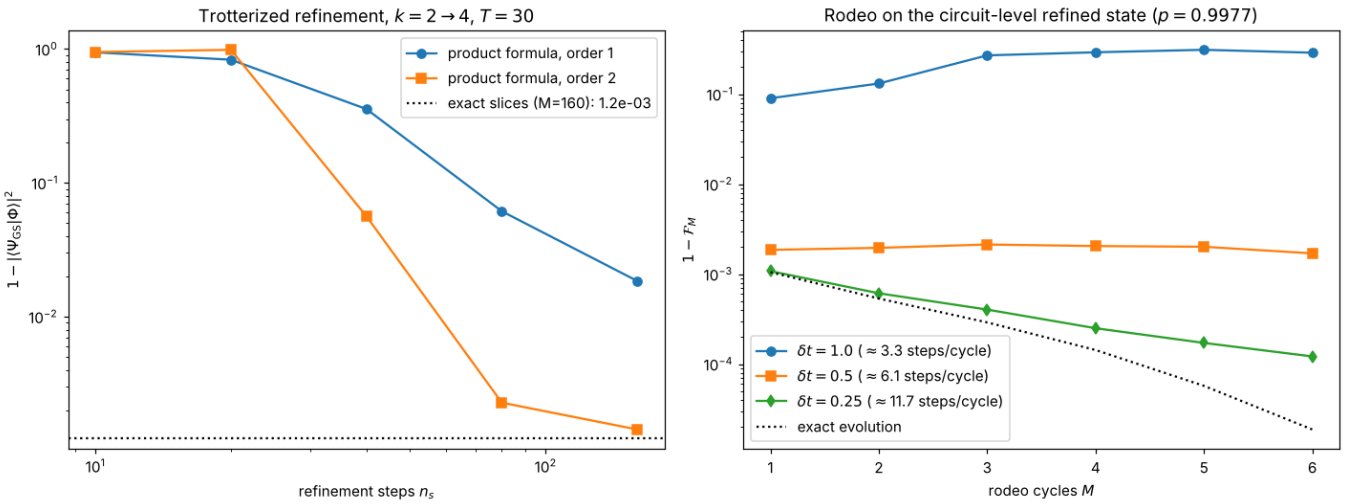


FIG. 6. The pipeline at the circuit level ($k = 4$, $N = 4$, $g = 1$). Left: infidelity of the Trotterized refinement versus the number of product-formula steps n_s along the $T = 30$ path, for first- and second-order formulas; the dotted line is the exact-slice reference. Right: rodeo infidelity versus the number of cycles M for a Trotterized controlled evolution with fixed step size δt (input: the $n_s = 80$ circuit-level refined state, $p = 0.9977$). Too-coarse steps are rejected by the filter itself (the $\delta t = 1$ acceptance collapses to 0.10); $\delta t = 0.25$ tracks the exact evolution down to the floor of the averaged post-selection map, 1.1×10^{-4}.

accepted runs, the acceptance P_M, the CNOTs of one attempt that runs all M cycles, $C_M^{\rm att} = C_{\rm prep} + \sum_{m\le M}\langle c_m\rangle$ — the coherent two-qubit depth proxy of a successful run — and the expected CNOTs per accepted preparation,

$$C_M = \frac{C_{\rm prep} + \sum_{m=1}^{M} \langle P_{m-1}\, c_m\rangle}{\langle P_M\rangle}, \qquad (12)$$

where $c_m = 2 \times 416\, n_m$ is the cost of the m-th controlled cycle, the expectation is over the random schedules, and the factor P_{m-1} inside it accounts for early termination of a run at the first ancilla outcome 1 (without early termination the numerator is simply $C_{\rm prep} + \sum_m \langle c_m\rangle$; the difference is 9% for the determinant at six cycles and below 2% for the inputs with $p > 0.99$). Rejected attempts add to the total work but not to the coherent depth of the accepted state; $C_M^{\rm att}$ is a full-length proxy for that depth, not the mean conditioned on acceptance. Equation (12) counts repeated state preparation only. One-time costs are reported separately and are not comparable across routes without a shot model: the UCCSD training took eight L-BFGS iterations in a noiseless simulation, each requiring the energy and its gradient — a hardware shot budget would need an estimator, a grouping scheme, a shot precision and a stopping criterion — and the energy search of the rodeo stage costs the scans of Sec. IV E, of order 10^3 four-cycle runs times their shots, with a preparation cost and peak statistics that depend on which input is scanned. For Q requested accepted states the natural total is $C_{\rm setup} + Q\, C_M$, and the setup is amortized away for large Q.

Figure 7 and Table II give the result. The first observation is that the high-overlap inputs approach a common product-formula-limited accuracy, $1-\mathcal{F} \simeq 1.1$–1.7×10^{-4} after six cycles, the floor of the averaged post-selection map at $\delta t = 0.25$ (Sec. V B); the determinant, which still carries substantial excited weight at $M = 6$ (1.8×10^{-3}), and the prolonged $k = 3$ state (7.6×10^{-4}) need more cycles to reach that regime, and six-cycle values are not measured asymptotes. The second observation is the ranking at *matched accuracy*, read off the minimum cost over inputs and cycle numbers, $M = 0$ included, for a target infidelity: for every target above the floor — 2×10^{-3}, 5×10^{-4} and 2×10^{-4} — the cheapest route is the unfiltered four-pair ansatz, 192 CNOTs at 1.8×10^{-4}, followed by the same state in its 1 312-CNOT template (and the adaptive ansatz of Sec. V D reaches the exact state with one more operator, 240 CNOTs); the cheapest route through the filter is the HF ramp, with 1.7×10^4, 4.0×10^4 and 6.4×10^4 CNOTs per accepted preparation after one, three and five cycles; and the cheapest refinement route, the $T^* = 16$ chain, costs 4.9×10^4, 7.4×10^4 and 9.4×10^4 after two, four and six cycles. Among these fixed inputs, refinement followed by the filter is tied within a factor 1.5 with filtering the unrefined determinant (1.8×10^{-3} after six cycles for 7.0×10^4 CNOTs), costs 2.9, 1.9 and 1.5 times more than the plain ramp at the three targets and 200–500 times more than the four-pair preparation, and the prolonged $k = 3$ state filtered directly (7.6×10^{-4} after six cycles, 6.7×10^4 CNOTs) is cheaper than refining it first. Comparisons at a fixed cycle number, or of the UCCSD *energy error* with the pipeline *infidelity*, would favor the pipeline by orders of magnitude; neither survives the comparison of like with like. Nor are these fixed operating points optimized minima: the step counts were chosen to reproduce the exact-slice overlap within 10^{-3} while the targets reach 2×10^{-4}, and paying

for a finer unfiltered evolution can be cheaper than paying for controlled filter cycles. Searching both adiabatic routes jointly over time, shift, step count and cycle number (Table V, Sec. V E) confirms this: at eight qubits the three targets are reached without any cycle by the ramp for 9.5×10^3, 1.9×10^4 and 2.5×10^4 CNOTs and by the refinement with $\mu_{\mathrm{buf}} = 3$ at $T = 6$ for 9.9×10^3, 1.4×10^4 and 2.0×10^4, a tie in which the shift-set gap of the projector path (3, against the ramp's 2) outweighs its coarse-term overhead (14 of 310 CNOTs per step).

Table S2 of the Supplemental Material [59] records the operating points behind the different fidelities quoted in this paper, which are not contradictory because they belong to different stages of idealization: the ideal filter of Sec. IV E and the digital pipeline of this section differ in refinement time, slice treatment, filter width and evolution, and the residual errors of the former are not achieved by the counted circuits of the latter.

What remains in favor of the refinement stage is qualitative but real. It contains no variational parameters: nothing is trained at full resolution, no expectation values are measured during state preparation, and the only classical inputs are the shift μ, the schedule T and the step count n_s — which, as noted in Sec. III C, were calibrated here against the exact answer rather than from coarse-space information. Its gate count is dominated by the refinement, which scales with the Trotter step of the fine Hamiltonian and with T^*, and is independent of how hard the variational landscape is. Whether this trade — trainability for depth — pays off is a question about regimes in which the full-space variational input is poor or expensive to train. For the pairing model with $N = 4$ that regime is not reached at $k = 4$ or $k = 6$, nor at $N = 6$ (Sec. S7 of the Supplemental Material [59]); with four pairs on eight levels it is (Sec. V E), and there both adiabatic routes beat the single-layer variational one, while the ramp baseline adds that, where an adiabatic stage is wanted, the shift-controlled path is at best tied with the plain ramp from the same determinant; we state the latter as the main negative finding of the present assessment.

D. The adaptive variational comparator

The steep UCCSD gate count is not irreducible, and much recent work aims precisely at shallower variational circuits (see Ref. [43] for a review). ADAPT-VQE [51] grows the ansatz one operator at a time, at each step appending the pool operator with the largest energy gradient $|\langle\psi|[H, K]|\psi\rangle|$ — an operator may be selected again — and re-optimizing all amplitudes. The pairing model makes the pool transparent: because the Hamiltonian conserves seniority and the reference is seniority-zero, the gradients of all singles and of all pair-breaking doubles vanish identically on the seniority-zero manifold, so the pool is either the $\frac{N}{2}(k - \frac{N}{2})$ occupied-pair→virtual-pair rotations of the restricted ansatz of Sec. V C or the *generalized* pool of all $k(k-1)/2$ pair-hopping generators $P_q^\dagger P_p$ − h.c., each a 48-CNOT block in the $2k$-qubit encoding. We ran ADAPT-VQE with both pools, in the seniority-zero pair basis of Sec. V E and with the exact gradient, for $(k, N) = (4, 4)$ and $(6, 4)$ at $g = 1$ and for $(8, 8)$ at $g = 1$ and 2 (Table III, Fig. 8). At $k = 4$ the restricted pool reproduces the four-pair ansatz exactly — four operators, 192 CNOTs, 1.78×10^{-4} — and the generalized pool adds the occupied–occupied pair rotation $(0, 1)$ as a fifth operator and reaches the exact ground state ($1 - F < 10^{-15}$, 240 CNOTs): the six-dimensional seniority-zero space is spanned. At sixteen qubits the adaptive ansatz does what the single-layer one cannot: with the generalized pool 19 operators (912 CNOTs) reach 1.5×10^{-4}, 23 (1 104 CNOTs) reach 1.7×10^{-5} and 27 (1 296 CNOTs) reach 4×10^{-6} at $g = 1$, and 25, 29 and 31 operators (1 200–1 488 CNOTs) reach 10^{-3}, 3×10^{-4} and 10^{-4} at $g = 2$, where the single-layer ansatz stops at 3×10^{-3} and 4×10^{-2}; the restricted pool with repeated operators is slower (29 operators for 10^{-4} at $g = 1$, no 3×10^{-4} within 40 at $g = 2$). Fed to the Trotterized filter of Sec. V E, the 1 104-CNOT state at $g = 1$ starts *below* the floor of the digital filter (1.7×10^{-5} against 4.8×10^{-5}) and the computed track degrades it to 4.8×10^{-5} in six cycles, while the 1 488-CNOT state at $g = 2$ (8.9×10^{-5}, above the floor 2.7×10^{-5}) improves by a factor three in six cycles at four hundred times its preparation cost (Table VI); either way the filter adds little to a state that already meets the targets. Qubit-ADAPT-VQE [52] would push further by drawing the pool from individual Pauli strings. The usual qualifications apply: the amplitudes are trained noise-free with exact gradients, each iteration measures |pool| gradients and re-optimizes the ansatz, and these one-time costs are excluded from the gate budget as they are for UCCSD; and the pool is the symmetry-adapted one that the seniority conservation of the pairing force makes available. A different route to a shallower circuit — keeping the Trotterized refinement path as the ansatz structure and optimizing a few of its layers, as in the optimized nucleation of Ref. [26] — is taken up in Sec. VI.

E. A larger model space: four pairs on eight levels

The sizes of Table II, and the $N = 4$ and $N = 6$ systems on $k = 4$ and 6 levels of Sec. S7 of the Supplemental Material [59] (where the ranking is the same), keep the variational input good and the filter cycles cheap. To test whether a larger and more correlated model space changes the ranking, we repeat the matched comparison for $N = 8$ particles on $k = 8$ levels — four pairs, sixteen qubits, the seniority-zero sector of $\binom{8}{4} = 70$ states inside the 12 870-dimensional $N = 8$ sector — for $g = 0.5$ to 2 (notebook, Sec. 20). Every stage of the pipeline conserves seniority (the Hamiltonian, the embedding, the path, the exactly exponentiated pair-hopping blocks and the filter), so the whole computation can be carried out

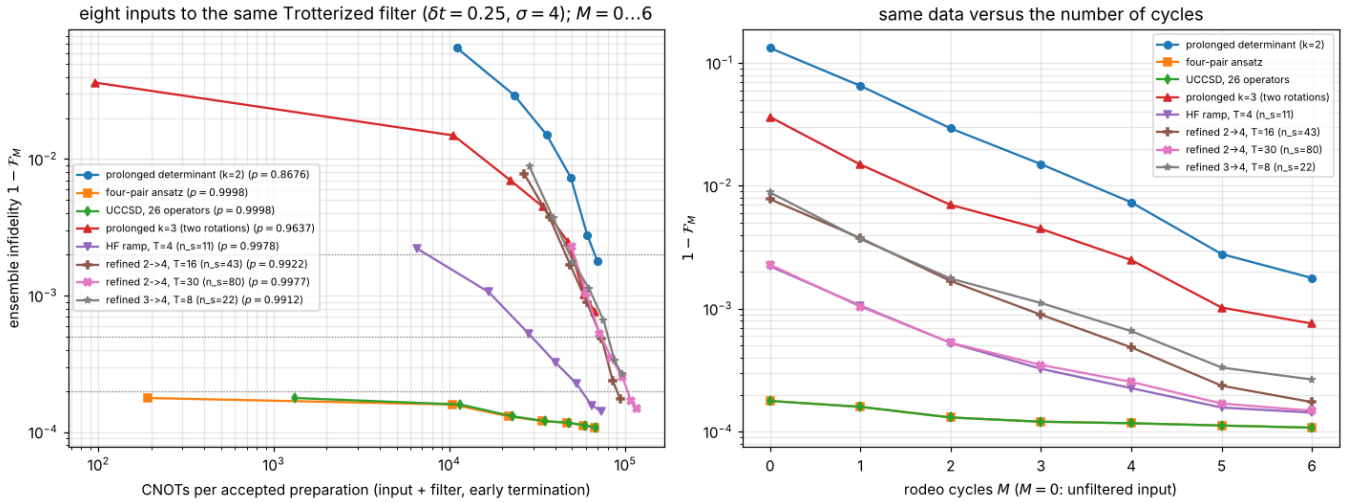


FIG. 7. Matched-observable comparison of eight inputs to the same Trotterized rodeo filter ($k = 4$, $N = 4$, $g = 1$; second order, $\delta t = 0.25$, $\sigma = 4$, 40 common Gaussian time sets): the prolonged $k = 2$ determinant, the four-pair restricted ansatz and the full UCCSD template (the same state at 192 and 1 312 CNOTs), the prolonged $k = 3$ coarse state, the HF adiabatic ramp, the Trotterized resolution-refined states at $T^* = 16$ and $T = 30$, and the $3 \to 4$ chain at $T = 8$ (refinement costs include the coarse term of Eq. (4)). Left: acceptance-weighted ensemble infidelity of the accepted runs versus CNOTs per accepted preparation, Eq. (12), as the number of cycles grows from $M = 0$ (unfiltered input; omitted for the gate-free determinant) to $M = 6$; dotted lines mark the targets 2×10^{-3}, 5×10^{-4} and 2×10^{-4} of the text. Right: the same infidelities versus M. The high-overlap inputs converge to the floor of the product-formula filter; the four-pair ansatz is below every target before any cycle.

TABLE II. Gate budget at $k = 4$ ($N = 4$, $g = 1$, eight system qubits) for eight fixed inputs to the same Trotterized rodeo filter ($\delta t = 0.25$, $\sigma = 4$, 40 sampled time sets); the minimum costs at the targets over the adiabatic controls are in Table V. p is the input overlap, $C_{\rm prep}$ the CNOT count of the input (refinement: $2\,n_s$ second-order half-steps of $296 + 14$ CNOTs for the $2 \to 4$ chains and of $296 + 350$ for the $3 \to 4$ chain, plus the coarse preparation; ramp: $2n_s \times 296$), $1 - \mathcal{F}_M$ the acceptance-weighted infidelity of the accepted runs, P_M the acceptance, $C_M^{\rm att}$ the CNOTs of one attempt running all M cycles and C_M the CNOTs per accepted preparation of Eq. (12). One-time training (variational inputs) and the energy scan are not included (see text). The UCCSD energy error of 1.4×10^{-3} (Table I) corresponds to the infidelity 1.8×10^{-4} listed here.

			$M = 0$	$M = 2$				$M = 6$			
input to the filter	p	$C_{\rm prep}$	$1 - \mathcal{F}_0$	$1 - \mathcal{F}_2$	P_2	$C_2^{\rm att}$	C_2	$1 - \mathcal{F}_6$	P_6	$C_6^{\rm att}$	C_6
prolonged determinant (k=2)	0.868	0	1.3×10^{-1}	2.9×10^{-2}	0.887	2.2×10^4	2.3×10^4	1.8×10^{-3}	0.854	6.6×10^4	7.0×10^4
four-pair ansatz (4 parameters)	0.99982	192	1.8×10^{-4}	1.3×10^{-4}	0.995	2.2×10^4	2.2×10^4	1.1×10^{-4}	0.986	6.6×10^4	6.7×10^4
UCCSD template (26 parameters)	0.99982	1 312	1.8×10^{-4}	1.3×10^{-4}	0.995	2.3×10^4	2.3×10^4	1.1×10^{-4}	0.986	6.7×10^4	6.8×10^4
prolonged k=3 state (96-CNOT prep.)	0.964	96	3.6×10^{-2}	7.0×10^{-3}	0.964	2.2×10^4	2.2×10^4	7.6×10^{-4}	0.948	6.6×10^4	6.7×10^4
HF adiabatic ramp, T=4 (n_s=11)	0.9978	6 512	2.2×10^{-3}	5.3×10^{-4}	0.994	2.8×10^4	2.8×10^4	1.4×10^{-4}	0.985	7.3×10^4	7.3×10^4
refined $2 \to 4$, T^*=16 (n_s=43)	0.9922	26 660	7.8×10^{-3}	1.7×10^{-3}	0.989	4.8×10^4	4.9×10^4	1.7×10^{-4}	0.979	9.3×10^4	9.4×10^4
refined $2 \to 4$, T=30 (n_s=80)	0.9977	49 600	2.3×10^{-3}	5.3×10^{-4}	0.994	7.1×10^4	7.2×10^4	1.5×10^{-4}	0.985	1.2×10^5	1.2×10^5
refined $3 \to 4$, T=8 (n_s=22)	0.9912	28 520	8.8×10^{-3}	1.8×10^{-3}	0.988	5.0×10^4	5.1×10^4	2.7×10^{-4}	0.978	9.5×10^4	9.6×10^4

in the pair basis, one bit per level, which we validated against the full-sector implementation at $k = 4$, $N = 4$ (Sec. S8 of the Supplemental Material [59]): the Hamiltonian and every Trotter term coincide with the seniority-zero blocks to machine precision and every anchor of this paper is reproduced. The gate counts remain those of the $2k$-qubit compilation of Sec. V: a first-order step costs 1 360 CNOTs at $k = 8$ (248 rotations, 1 856 CNOTs controlled), the coarse term of the $4 \to 8$ chain is a phase conditioned on the eight appended qubits, $2^8 - 2 = 254$ CNOTs with the Gray-code convention that gave 14 for the $2 \to 4$ chain, and the coarse term of a $6 \to 8$ chain is the $k = 6$ step with every rotation conditioned on four qubits, 4 886 CNOTs (ancilla-free convention; 1 028 with a shared flag, Sec. V B). The one-qubit-per-level pair encoding, available because the model conserves seniority, is quoted alongside: there a step costs 112 CNOTs, a controlled step 240, a pair-double rotation 4 and the $4 \to 8$ coarse term 14.

a. What changes with the size. Table S5 of the Supplemental Material [59] lists the ingredients. The HF overlap falls from 0.868 to 0.690 at $g = 1$ and to 0.25 at $g = 2$, the correlation energy grows to 1.1 and 5.5, and the restricted pair-double ansatz — now 16 occupied-

TABLE III. ADAPT-VQE in the seniority-zero pair basis (gradient tolerance 10^{-6}, at most 40 operators; 48 CNOTs per operator in the $2k$-qubit encoding, 4 in the pair encoding): the pool size and the number of operators (CNOTs) at which the infidelity first falls below 10^{-3}, 3×10^{-4}, 10^{-4} and 10^{-5}; a dash means not within 40 operators; "gen." is the generalized pool and the pool size is given in parentheses. The last column gives the infidelity of the single-layer restricted ansatz used in Tables II and VI.

(k,N,g)	pool	10^{-3}	3×10^{-4}	10^{-4}	10^{-5}	1 layer
$(4,4,1)$	pairs (4)	4 (192)	4 (192)	—	—	1.8×10^{-4}
$(4,4,1)$	gen. (6)	4 (192)	4 (192)	5 (240)	5 (240)	
$(6,4,1)$	pairs (8)	8 (384)	9 (432)	11 (528)	13 (624)	7.6×10^{-4}
$(6,4,1)$	gen. (15)	8 (384)	9 (432)	9 (432)	9 (432)	
$(8,8,1)$	pairs (16)	17 (816)	28 (1344)	29 (1392)	—	3.1×10^{-3}
$(8,8,1)$	gen. (28)	19 (912)	19 (912)	23 (1104)	27 (1296)	
$(8,8,2)$	pairs (16)	37 (1776)	—	—	—	4.2×10^{-2}
$(8,8,2)$	gen. (28)	25 (1200)	29 (1392)	31 (1488)	—	

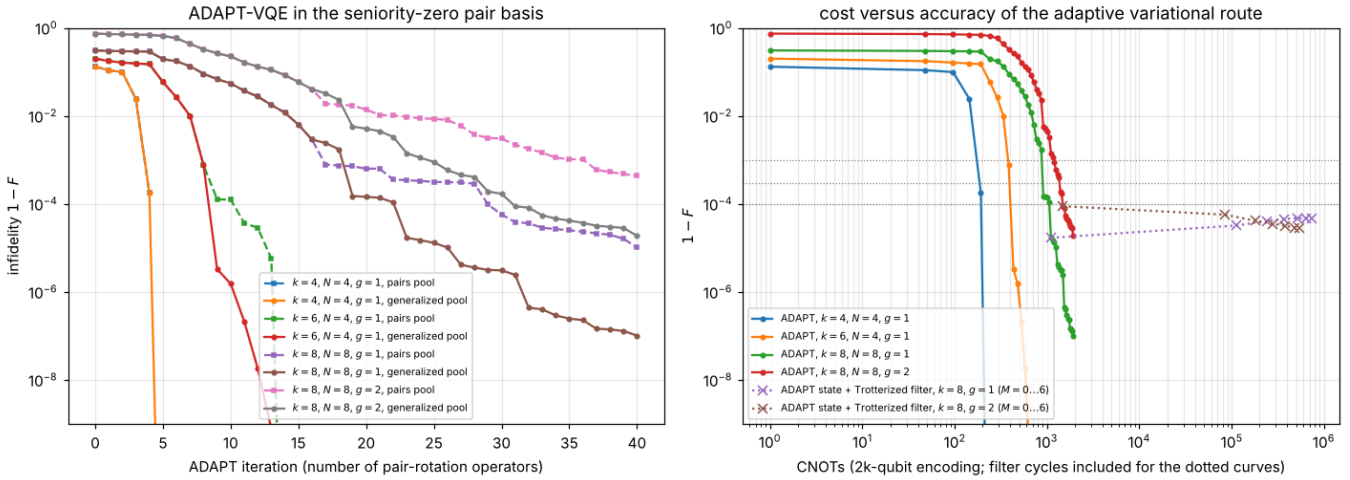


FIG. 8. ADAPT-VQE in the seniority-zero pair basis. Left: infidelity versus the number of operators for the restricted (dashed) and the generalized (solid) pool at $(k,N)=(4,4)$, $(6,4)$ and $(8,8)$. Right: infidelity against CNOTs for the generalized pool; the dotted curves feed the $(8,8)$ ADAPT states at the first iteration below 10^{-4} to the Trotterized filter of Sec. V E ($M=0$ to 6), which degrades the $g=1$ input, already below its floor, and improves the $g=2$ input by a factor three.

pair→virtual-pair rotations, 768 CNOTs — degrades from the 1.8×10^{-4} of $k=4$ to 3.1×10^{-3} at $g=1$ and 4.2×10^{-2} at $g=2$: the single-reference variational input is no longer good. For an exact coarse eigenstate and a Galerkin pair of Hamiltonians, $P^\dagger H_{\rm high}P=H_{\rm low}$, the response norm of the projector path at its start is

$$\chi_u(0)=\frac{\|Q\,H_{\rm high}P\,|\psi_{\rm low}\rangle\|^2}{b^2},\qquad Q=\mathbb{1}-PP^\dagger,\quad(13)$$

since $\partial_u H=B-A$ does not couple the tracked state to the other coarse states and the null space of $P^\dagger$ sits at energy b above it; for the filled coarse seed of the pairing model the numerator counts pair transfers, $\chi_u(0)=P(k-P)\,g^2/(4b^2)$ with $P=N/2$ (verified to 10^{-9} for every g; g^2/b^2 at $k=4$, $N=4$). At the shift $\mu_{\rm buf}=0.6$ of Sec. IV B it is four times larger than at $k=4$, and the refinement time grows, at matched target, from $T^*=40$ to 50 for $1-p<10^{-3}$ and from 16 to 30 for 10^{-2}. The shift is the remedy. Table IV scans $\mu_{\rm buf}$ from 0.6 to 16: T^* falls to 8 at $\mu_{\rm buf}=2$ and 6 at $\mu_{\rm buf}=4$ for $g=1$, and to 8 at $\mu_{\rm buf}=8$ for $g=2$, while the number of second-order steps that reproduce the exact-slice overlap first falls with the shift and then rises again (the generator norm grows with μ), with a minimum of 40 steps for $g=1$ and 80 for $g=2$. At the well-chosen shift all chains $k_{\rm low}\to8$ with $k_{\rm low}=4$ to 7 collapse onto one curve (Fig. S8 of the Supplemental Material [59], left): the transport is controlled by the shift-set gap, not by the starting overlap, and a better coarse state buys nothing once $\mu_{\rm buf}\gtrsim2$. The HF ramp from the same determinant, $H(s)=D+\sin^2(\pi s/2)\sum_{pq}h_{pq}$, reaches $1-p=7\times10^{-5}$ at $T=6$, its minimum gap is 2δ at every coupling and, empirically, its transport time does not grow with g over the range studied (Fig. S8 of the Supplemental Material [59], right) — the gap alone does not establish this, since the derivative couplings matter as well — whereas the refinement at fixed shift slows down ($\chi_u(0)\propto g^2$) and must be re-tuned.

TABLE IV. Shift scan of the $4 \to 8$ chain at $k = 8$, $N = 8$: the response norm $\chi_u(0) = P(k-P)g^2/(4b^2)$, the refinement time T^* for $1 - p < 10^{-3}$ (exact slices, grid $2, 4, 6, 8, 10, 12, 16, 20, 25, 30, \ldots$), the second-order steps n_s reproducing the exact-slice overlap within 10^{-3}, and the refinement cost $2n_s(1\,360 + 254)$ CNOTs (a dash: not Trotterized because $T^* > 25$).

	$g = 1$				$g = 2$			
$\mu_{\rm buf}$	$\chi_u(0)$	T^*	n_s	CNOTs	$\chi_u(0)$	T^*	n_s	CNOTs
0.6	11.1	50	—	—	44.4	> 64	—	—
1	4.00	25	160	5.2×10^5	16.0	64	—	—
2	1.00	8	40	1.3×10^5	4.00	25	320	1.0×10^6
4	0.25	6	40	1.3×10^5	1.00	12	160	5.2×10^5
8	0.063	8	40	1.3×10^5	0.25	8	80	2.6×10^5
16	0.016	10	80	2.6×10^5	0.063	8	80	2.6×10^5

b. The digital filter at these norms. The Hamiltonian norm is five times larger than at $k = 4$, and the fixed-step filter of Sec. V B does not survive the $\delta t = 0.25$ operating point: the averaged post-selection map $\mathcal{K}$ at $\sigma = 4$ (the width that keeps $\sigma\Delta \simeq 10$ as before) has dominant eigenvalue 0.79 — the acceptance decays by a fifth per cycle — and a floor of 2.3×10^{-3}, above every target; this is the diagnostic rejection of Fig. 6 in a stronger form (an earlier, unconverged quadrature had suggested an eigenoperator with zero ground-state overlap; the converged map keeps 0.998 of the ground state). At $\delta t = 0.1$ the floor is 4.8×10^{-5} with eigenvalue 0.9932 (acceptance decaying by 0.7% per cycle), at $\delta t = 0.05$ it is 3.1×10^{-6}. We operate at $\delta t = 0.1$, where a cycle costs on average 32 controlled second-order steps, 1.2×10^5 CNOTs — eleven times the $k = 4$ cycle, which is the single most important change: a filter cycle now costs as much as doubling the step count of the adiabatic stage. At $g = 2$ the wider spectrum requires $\sigma = 1.5$ ($\sigma\Delta = 9$) and $\delta t = 0.05$ (floor 2.7×10^{-5}; 4.0×10^{-4} at 0.1), 9.0×10^4 CNOTs per cycle.

c. Matched comparison. Table VI repeats the fixed-input comparison of Sec. V C with eleven inputs at $g = 1$ and seven at $g = 2$ (forty common time sets, $M = 0$ to 6, bootstrap standard errors below 10% for every input with $p > 0.99$). Table V and Fig. 9 add the operating-point search that fixed inputs cannot give: the ensemble of accepted runs is evaluated exactly as $\mathcal{K}^M(\rho_{\rm in})/\operatorname{Tr}\mathcal{K}^M(\rho_{\rm in})$ with the averaged map of Sec. V B (independent cycle times make the cost of Eq. (12) deterministic as well, so no sampling error enters), the two adiabatic routes are searched jointly over time, shift, step count and cycle number with the same grid density, $M = 0$ included, and the adaptive ansatz is admitted at every prefix; the sampled values of Table VI agree with the exact ones within their bootstrap errors.

Three things have changed relative to $k = 4$ among the fixed-ansatz and adiabatic routes; the adaptive ansatz, discussed last, stands apart. First, the unfiltered single-layer preparation no longer meets any target: the 16-pair ansatz needs two, four and six cycles for 10^{-3}, 3×10^{-4} and 10^{-4}, at 2.4×10^5, 4.9×10^5 and 7.4×10^5 CNOTs per accepted preparation, and at $g = 2$ six cycles for 10^{-3} without reaching 3×10^{-4}; both adiabatic routes are now cheaper than it by factors 2–3, where at $k = 4$ they were 50–130 times dearer than the four-pair ansatz. Second, filtering the bare determinant is no longer an option: from $p = 0.69$ six cycles reach only 7×10^{-3} (8.4×10^5 CNOTs, acceptance 0.67), from $p = 0.25$ at $g = 2$ only 4×10^{-2}. Third — and this is where the joint search matters — the cheapest adiabatic operating points at $g = 1$ use *no filter cycle*: a cycle costs 1.2×10^5 CNOTs, as much as doubling the step count of a $T = 6$ sweep, and the finer digital evolution buys more. The ramp reaches the three targets at $T = 6$ with 40, 60 and 80 steps for 1.1×10^5, 1.6×10^5 and 2.2×10^5 CNOTs, the refinement with $\mu_{\rm buf} = 4$ at the same T and step counts for 1.3×10^5, 1.9×10^5 and 2.6×10^5: once the shift is of order the pair-coupling scale $g\sqrt{P(k-P)}/2$ the two routes travel at the same speed, and the 19% between them is exactly the conditioned phase of the coarse term (254 of 1 614 CNOTs per step). At $g = 2$ the ramp needs cycles for the two tighter targets (4.0×10^5 and 5.8×10^5 CNOTs) while the refinement at $\mu_{\rm buf} = 16$ reaches 3×10^{-4} unfiltered and 10^{-4} with two cycles (3.9×10^5 and 5.7×10^5): a tie within 3%, with the refinement 19% dearer at 10^{-3}. At $k = 4$ the same search turns the 1.5–2.9 margin of the fixed inputs of Table II into a tie as well: the ramp reaches the targets for 9.5×10^3, 1.9×10^4 and 2.5×10^4 CNOTs and the refinement with $\mu_{\rm buf} = 3$ (a shift-set gap of 3 against the ramp's 2) at $T = 6$ for 9.9×10^3, 1.4×10^4 and 2.0×10^4, again without any cycle. A nontrivial coarse stage does not help: the $6 \to 8$ chain from the eight-parameter coarse ansatz starts from $p_0 = 0.926$ but its coarse term costs 4 886 CNOTs per application in the ancilla-free compilation, so the chain costs 10^6 CNOTs before any cycle (3.8×10^5 with the shared-flag compilation of Sec. V B, still above the ramp), and the prolonged $k = 6$ state filtered directly reaches 1.4×10^{-3} after six cycles. In the pair encoding every adiabatic cost drops by a factor 12 and every cycle by 8, and the ranking is unchanged.

The adaptive variational ansatz of Sec. V D changes the scale of the whole table: 19 operators (912 CNOTs) reach 1.5×10^{-4} and 23 (1 104 CNOTs) 1.7×10^{-5} at $g = 1$, and 25–31 operators (1 200–1 488 CNOTs) reach the three targets at $g = 2$, without any filter cycle, two orders of magnitude below every other route at every target; the $g = 1$ state lies below the floor of the digital filter and the $g = 2$ state within a factor three of it, so the filter has little to add to them. Among the routes that need no classical optimization the ramp is the cheapest at $g = 1$ and tied with the refinement at $g = 2$; among all routes the adaptive ansatz is, at both sizes and both couplings.

The answer to the question that motivates this subsection is therefore split three ways. A larger and more correlated space favors the adiabatic routes, refinement

TABLE V. Minimum CNOTs per accepted preparation reaching the target infidelity, from the exact averaged channel of the digital filter, per route. The adiabatic routes are searched jointly over the time T, the second-order step count n_s and the cycle number $M = 0, \ldots, 6$ (ramp: $T \in \{4, 6, 8, 12\}$ at $k = 4$, $\{4, 6, 8\}$ and $\{8, 12, 16\}$ at $k = 8$ for $g = 1$ and 2; refinement: in addition the shift $\mu_{\mathrm{buf}} \in \{0.6, 1.5, 3\}$ at $k = 4$, $\{2, 4, 8\}$ and $\{4, 8, 16\}$ at $k = 8$, with $T \in \{6, 8, 12, 16, 30\}$, $\{6, 8, 12\}$ and $\{8, 12, 16\}$; n_s from 8 to 128 at $k = 4$ and 20 to 320 at $k = 8$), the fixed inputs over M, and ADAPT over its prefixes. The operating point is given in parentheses as (T, n_s, M), $(\mu_{\mathrm{buf}}, T, n_s, M)$ or (operators); filter settings as in Tables II and VI; $2k$-qubit encoding, coarse terms in the ancilla-free convention; a dash: not reached within six cycles.

route	target infidelity		
$k = 4$, $N = 4$, $g = 1$ ($\delta t = 0.25$, $\sigma = 4$, cycle 1.1×10^4 CNOTs)			
	2×10^{-3}	5×10^{-4}	2×10^{-4}
ADAPT, generalized pool	240 (5)	240 (5)	240 (5)
four-pair ansatz	192 (M=0)	192 (M=0)	192 (M=0)
HF ramp	9.5×10^3 (4, 16, 0)	1.9×10^4 (6, 32, 0)	2.5×10^4 (8, 43, 0)
refinement $2 \to 4$	9.9×10^3 (3, 6, 16, 0)	1.4×10^4 (3, 6, 22, 0)	2.0×10^4 (3, 6, 32, 0)
refinement $3 \to 4$, $T = 8$	6.2×10^4 (22, M=3)	8.5×10^4 (22, M=5)	—
prolonged k=3 state + filter	5.7×10^4 (M=5)	—	—
determinant + filter	—	—	—
$k = 8$, $N = 8$, $g = 1$ ($\delta t = 0.1$, $\sigma = 4$, cycle 1.2×10^5 CNOTs)			
	10^{-3}	3×10^{-4}	10^{-4}
ADAPT, generalized pool	912 (19)	912 (19)	1 104 (23)
16-pair ansatz + filter	2.4×10^5 (M=2)	4.9×10^5 (M=4)	7.4×10^5 (M=6)
HF ramp	1.1×10^5 (6, 40, 0)	1.6×10^5 (6, 60, 0)	2.2×10^5 (6, 80, 0)
refinement $4 \to 8$	1.3×10^5 (4, 6, 40, 0)	1.9×10^5 (4, 6, 60, 0)	2.6×10^5 (4, 6, 80, 0)
refinement $6 \to 8$ (coarse ansatz)	1.1×10^6 (M=1)	1.4×10^6 (M=3)	1.7×10^6 (M=5)
prolonged k=6 ansatz + filter	—	—	—
determinant + filter	—	—	—
$k = 8$, $N = 8$, $g = 2$ ($\delta t = 0.05$, $\sigma = 1.5$, cycle 9.1×10^4 CNOTs)			
	10^{-3}	3×10^{-4}	10^{-4}
ADAPT, generalized pool	1 200 (25)	1 392 (29)	1 488 (31)
16-pair ansatz + filter	5.6×10^5 (M=6)	—	—
HF ramp	2.2×10^5 (8, 80, 0)	4.0×10^5 (8, 80, 2)	5.8×10^5 (8, 80, 4)
refinement $4 \to 8$	2.6×10^5 (8, 8, 80, 0)	3.9×10^5 (16, 8, 120, 0)	5.7×10^5 (16, 8, 120, 2)
prolonged k=6 ansatz + filter	—	—	—
determinant + filter	—	—	—

included, over a *fixed* single-layer variational preparation and over filtering the bare determinant, because the single-reference ansatz degrades and the cycles become expensive while the adiabatic transport, properly shifted, does not slow down. It does not favor them over an *adaptively* grown ansatz, whose seniority-zero pool is small and exactly the right one. And it does not separate the shift-controlled projector path from the ordinary ramp: once the shift is tuned the two differ by the coarse-term overhead, in the ramp's favor at $g = 1$ and within 3% at $g = 2$; the specific asset of resolution refinement — a coarse eigenstate better than the determinant — is not exercised by a hierarchy seeded by the determinant, and a better coarse state costs, with the ancilla-free compilation used here, more to embed than it saves. The shift must be chosen with the size: keeping $\chi_u(0)$ of Eq. (13) of order unity or below reproduces the optimum found here, and the $\mu_{\mathrm{buf}} = 0.6$ of Sec. IV B would make the $k = 8$ refinement six times slower than necessary.

VI. DISCUSSION AND OUTLOOK

What this work delivers is, first, a reference implementation: the complete refinement-plus-filter chain with every stage benchmarked against exact results — a gate-level UCCSD coarse solver, a gate-free prolongation with a counted conditioned coarse term, a gapped adiabatic refinement, and a rodeo filter whose ensemble infidelity follows Eq. (10) — all validated a second time at the circuit level with product formulas, at eight and at sixteen qubits. The two algorithms are complementary in the intended sense: refinement raises the overlap from 0.87 to 0.998, and the filter removes the adiabatic tail geometrically. The register and the circuits of all stages — prolongation, the adiabatic path, the shift rule and the rodeo cycle — are written for the generic $2k$-qubit register and do not exploit seniority for compression; the favorable dynamics and the small variational inputs available here do rely on the seniority conservation of the pairing force.

Second, three results that do not depend on the model. The response norm of the refinement path at its start,

TABLE VI. Gate budget at $k = 8$, $N = 8$ (sixteen qubits) for the inputs to the same Trotterized rodeo filter (second order, $\sigma = 4$ and $\delta t = 0.1$ at $g = 1$, $\sigma = 1.5$ and $\delta t = 0.05$ at $g = 2$, forty time sets, $E_{\rm obj} = E_0$), in the $2k$-qubit encoding of Sec. V. Columns as in Table II ($C_M^{\rm att}$, P_M and C_M omitted; the minimum costs at the targets, from the exact averaged channel and a joint search over the adiabatic controls, are in Table V). Refinement costs include the coarse term (254 CNOTs per application for $4 \to 8$, 4 886 for $6 \to 8$) and, for the $6 \to 8$ chain, the 384-CNOT coarse ansatz.

input to the filter	p	$C_{\rm prep}$	$1 - \mathcal{F}_0$	$1 - \mathcal{F}_2$	$1 - \mathcal{F}_6$
$g = 1$ ($E_0 = 8.88917$, cycle 1.2×10^5 CNOTs)					
HF determinant	0.690	0	3.1×10^{-1}	9.7×10^{-2}	1.0×10^{-2}
16-pair ansatz	0.9969	768	3.1×10^{-3}	7.2×10^{-4}	9.8×10^{-5}
prolonged k=6 ansatz	0.926	384	7.4×10^{-2}	1.8×10^{-2}	1.8×10^{-3}
HF ramp, T=4 (n_s=20)	0.9960	54 400	4.0×10^{-3}	1.0×10^{-3}	1.4×10^{-4}
HF ramp, T=6 (n_s=40)	0.9996	108 800	3.5×10^{-4}	1.2×10^{-4}	5.3×10^{-5}
ref. 4→8, $\mu_{\rm buf}$=1, T=25 (n_s=160)	0.9994	516 480	5.7×10^{-4}	1.7×10^{-4}	5.9×10^{-5}
ref. 4→8, $\mu_{\rm buf}$=2, T=8 (n_s=40)	0.9987	129 120	1.3×10^{-3}	3.4×10^{-4}	6.9×10^{-5}
ref. 4→8, $\mu_{\rm buf}$=2, T=10 (n_s=80)	0.9997	258 240	3.3×10^{-4}	1.1×10^{-4}	5.1×10^{-5}
ref. 4→8, $\mu_{\rm buf}$=4, T=6 (n_s=40)	0.9996	129 120	3.5×10^{-4}	1.2×10^{-4}	5.3×10^{-5}
ref. 6→8 (coarse ansatz), $\mu_{\rm buf}$=2, T=8 (n_s=80)	0.9987	999 744	1.3×10^{-3}	3.3×10^{-4}	7.1×10^{-5}
ADAPT, generalized pool (23 op.)	0.99998	1 104	1.7×10^{-5}	4.1×10^{-5}	4.8×10^{-5}
$g = 2$ ($E_0 = 2.48659$, cycle 9.0×10^4 CNOTs)					
HF determinant	0.253	0	7.5×10^{-1}	3.8×10^{-1}	3.1×10^{-2}
16-pair ansatz	0.958	768	4.2×10^{-2}	8.8×10^{-3}	6.9×10^{-4}
HF ramp, T=8 (n_s=80)	0.9992	217 600	8.1×10^{-4}	1.9×10^{-4}	3.5×10^{-5}
HF ramp, T=12 (n_s=160)	0.9998	435 200	2.3×10^{-4}	7.0×10^{-5}	2.9×10^{-5}
ref. 4→8, $\mu_{\rm buf}$=2, T=25 (n_s=320)	0.9997	1 032 960	2.9×10^{-4}	8.4×10^{-5}	3.0×10^{-5}
ref. 4→8, $\mu_{\rm buf}$=8, T=8 (n_s=80)	0.9992	258 240	7.6×10^{-4}	1.8×10^{-4}	3.5×10^{-5}
ADAPT, generalized pool (31 op.)	0.99991	1 488	8.9×10^{-5}	4.2×10^{-5}	2.8×10^{-5}

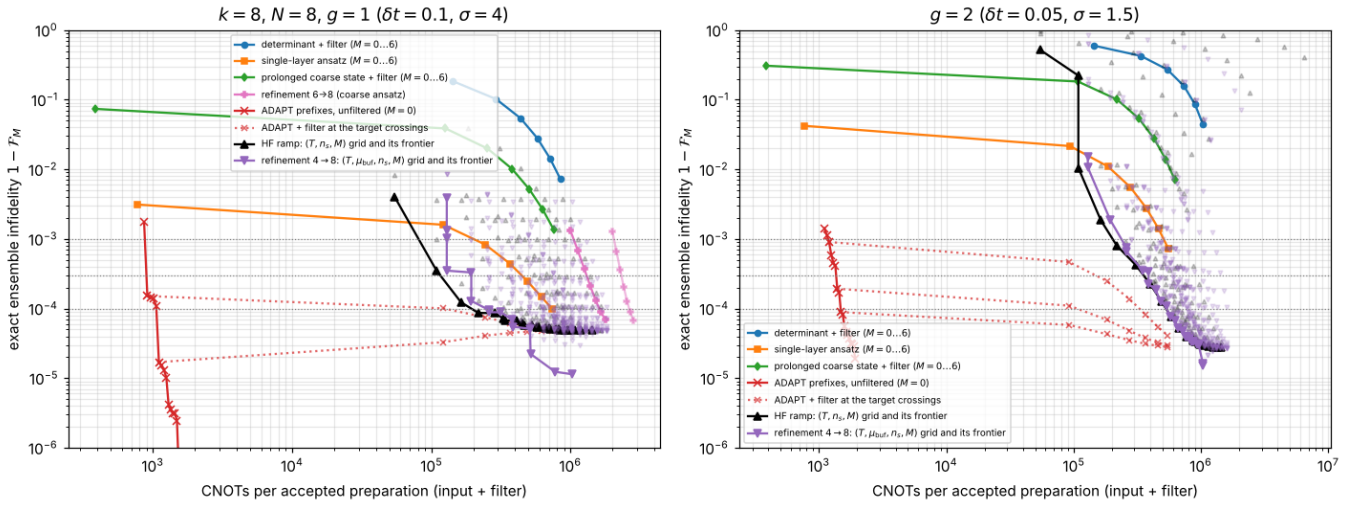


FIG. 9. Matched-observable comparison at $k = 8$, $N = 8$ with the exact averaged channel of the digital filter: ensemble infidelity of the accepted runs against CNOTs per accepted preparation, Eq. (12), at $g = 1$ (left; $\delta t = 0.1$, $\sigma = 4$) and $g = 2$ (right; $\delta t = 0.05$, $\sigma = 1.5$). Lines with markers: the fixed inputs of Table VI as M grows from 0 to 6, the ADAPT prefixes unfiltered (crosses) and the three prefixes at the target crossings through the filter (dotted); faint symbols: every point of the (T, n_s, M) grid of the ramp and of the $(\mu_{\rm buf}, T, n_s, M)$ grid of the $4 \to 8$ refinement, with their lower envelopes drawn (the minima of Table V). Dotted lines mark the targets 10^{-3}, 3×10^{-4} and 10^{-4}.

Eq. (13), says how the shift must grow with the size and the coupling for the transport to keep its speed — for the pairing seed it is $P(k - P)g^2/(4b^2)$, and the shift that is right at $k = 4$ is six times too slow at $k = 8$ — and shows why a scan of the refinement time against the shift-controlled gap cannot identify a scaling law, since the scan varies the quantity that the inverse-gap argument assumes bounded; the pair-counting coefficient is specific to this seed, the identity is not, and keeping $\chi_u(0)$ of order one is a tested shift rule rather than a theorem that fixes the transport time. The digital filter's long-run state and acceptance are the dominant eigenopera-

tor and eigenvalue of the averaged post-selection map, which sets the accuracy floor (1.1×10^{-4} at $\delta t = 0.25$ for eight qubits, 4.8×10^{-5} at $\delta t = 0.1$ for sixteen) and the slow geometric decay of the acceptance, and which, when the step is too coarse for the Hamiltonian norm, puts the floor above every target while the acceptance decays by a fifth per cycle, so that the filter rejects the corrupted evolution — a useful diagnostic, not a guarantee against smaller coherent errors; the map must be integrated piecewise between the step-count boundaries to be trusted. And the rigorous sufficient conditions on the adiabatic time and the Trotter step count overestimate the resources actually needed by two to three orders of magnitude, for an identifiable reason: they pay for the global norm and gap where the dynamics pays only for the couplings of the tracked state.

Third, and this is the finding the benchmark was built for, the ranking of the routes. Compared on the same observable — the ground-state infidelity — with the same Trotterized filter and with the adiabatic controls searched jointly, refinement is at $N = 4$ on four levels tied with an ordinary adiabatic ramp from the Hartree–Fock Hamiltonian (both unfiltered, within 30% either way), beats filtering the unrefined determinant, and is beaten by two orders of magnitude by the four-parameter pair-rotation ansatz, which needs no filter at all (Tables II and V, Sec. S7 of the Supplemental Material [59]). At four pairs on eight levels (Sec. V E) the single-layer ansatz is no longer good and the cycles are eleven times dearer, so both adiabatic routes overtake it and the filtered determinant by factors 2–3 and more; the ramp stays ahead of refinement by the cost of the conditioned coarse term at $g = 1$ and ties with it at $g = 2$, once the shift is scaled with the size; and an ADAPT-VQE ansatz grown from the seniority-zero pool (Sec. V D) reaches 10^{-4} with 1 100–1 500 CNOTs at both couplings, two orders of magnitude below every other route. The reason is simple, and Sec. V C states it as the ideal large-gap law $1 - \mathcal{F}_M \simeq \frac{1-p}{p} 2^{-M}$: a better input saves cycles only logarithmically in its odds ratio, so whether a preparation stage repays its gates depends on the price of a cycle relative to the price of the stage, and the size dependence of the verdict follows from the size dependence of that ratio. What the refinement offers in exchange is that it is optimizer-free and free of training measurements during preparation, and that its cost is set by the Hamiltonian evolution rather than by the trainability of a landscape. Whether that trade pays off against a fixed variational preparation is, as Sec. V E shows, a question of size and coupling, and the answer turns in favor of the adiabatic routes already at sixteen qubits; against an adaptive one it does not pay off at any size studied, as long as a small, symmetry-adapted pool exists and its gradients can be measured; whether the *shift-controlled* path pays off against the plain ramp is a question about coarse eigenstates that are better than the determinant, which a hierarchy seeded by the determinant cannot answer, and about hardware on which the training measurements of a variational loop dominate. We regard the present work as the controlled baseline against which such claims should be measured, not as evidence for them.

a. Limitations. Four limitations bound the claims. The comparison is at the level of coherent two-qubit gate counts for an ideal device; no noise model enters, and the acceptance statistics of the filter, which diagnose a corrupted evolution, are the only place where an imperfect device is visible. The costs are per accepted preparation and exclude the one-time costs — the training measurements of the variational inputs, the energy scan of the filter, and the calibration of the refinement controls — which have no common shot model and are amortized over the number of states requested. The refinement controls (T, μ, n_s), the filter step δt and its target energy are calibrated here against the exact answer; an implementable protocol would set μ from the coarse spectrum and the closed-form response norm, read T and n_s from the convergence of the measured energy, and locate the target by the scan, with the acceptance as the check, and we have not demonstrated it. And the adaptive comparator is trained with exact gradients on a symmetry-adapted pool that the pairing force makes available; a pool that must include seniority-breaking operators, or gradients estimated from shots, would raise its one-time cost and may change its convergence.

b. Relation to symmetry-restoration and subspace approaches. The results invite a comparison with the pairing-model work of Lacroix and collaborators summarized in the Introduction. Both programs start from a cheap, imperfect trial state and sharpen it; the difference is what is sharpened and how. In the symmetry-restoration route the trial state breaks a symmetry on purpose and the projector — phase estimation on the symmetry operator [27], an oracle [28] or a measurement-side projection [30] — removes the unwanted sectors, after which a Krylov or phase-estimation stage refines the energy [31]. Our coarse state conserves particle number and spin by construction; what it lacks is *resolution*, and the refinement supplies it along a gapped path while the rodeo cycle filters in *energy* rather than in a symmetry quantum number. The two filters are structurally the same resource — one ancilla, a controlled unitary and a post-selected measurement — and they compose: a symmetry-projected input would enter our pipeline exactly as the UCCSD state does. The quantum subspace methods of Refs. [35–37] are complementary in a different way: they trade coherent depth for measurements and a classical generalized eigenvalue problem, and their accuracy is bounded by the quality of the states that span the subspace, whereas the rodeo filter converges geometrically in depth but requires the controlled evolution. Two observations of that work bear on ours, with the limits of an energy filter kept in mind. Ref. [9] documents that ADAPT-VQE with a symmetry-breaking pool can collapse to unphysical energies on degenerate pairing systems. A downstream energy filter can repair such an input only insofar as the input retains a nonzero overlap

with the target: it cannot recover a state of zero overlap, cannot move weight between particle-number sectors that the evolution conserves, and cannot separate exactly degenerate states by energy alone, so a symmetry-broken ADAPT state would have to be symmetry-projected before, not instead of, the filter. Ref. [37] shows that the accuracy of the one-body Green's function of a small superfluid system, computed with the pure pairing Hamiltonian across the normal-to-superfluid transition driven by the pairing strength, is controlled by the quality of the even-N ground state; a filtered state of the kind prepared here is a possible input to that construction. The compact pair-to-qubit encoding used there [27, 37] is available precisely because the pairing force conserves seniority; a seniority-breaking particle-hole term, which is not part of the model of Ref. [37], is the ingredient that would force one out of it and onto the full $2k$-qubit register used throughout this work.

c. Relation to nucleation. The nucleation method of Ref. [26] is the closest relative of the refinement stage, and the comparison is instructive in both directions. Both start from a small problem solved exactly on a classical computer and compiled into a circuit, and both reach the target through a sequence of Trotterized adiabatic interpolations with the $\sin^2(\pi s/2)$ schedule of Eq. (3), each of which changes the Hamiltonian only locally: nucleation switches on the couplings of newly added lattice sites, initialized in $|+\rangle$, to the existing bulk; refinement switches off the conditioned coarse term A of Eq. (4) and switches on the matrix elements of $H_{\rm high}$ that connect the occupied coarse levels to the appended ones, initialized empty in $|0\rangle$. Nucleation grows the domain at fixed resolution, refinement the resolution at fixed domain, and the two compose in principle: a lattice or shell-model problem could be nucleated to its physical size in a coarse basis and then refined to the full basis, or the reverse, with the same Trotterized machinery and the same filter at the end.

What Ref. [26] adds to the present assessment is twofold. First, the sufficient conditions of Sec. III C, which overestimate the resources of a twelve-site Ising nucleation by orders of magnitude ($T \simeq 40$ and 2×10^3 Trotter steps suffice against bounds of 2.6×10^3 and 2×10^{10}) and those of the refinement path by the factors 3×10^2 and 2×10^2 of Sec. IV D; the distance between bound and practice is the room in which the inverse-gap scaling of Ref. [25] operates. Second, and more directly relevant to our negative finding, Ref. [26] introduces *optimized nucleation*: the Trotterized growth circuit is kept as the ansatz structure and its Trotter angles as the initial parameters, and a small number L of layers per stage — far below the Trotter step count of the non-variational version — is then optimized layer by layer to minimize the energy. On a 5×5 transverse-field Ising model this reaches a lower energy error at fewer two-qubit gates than converting a matrix-product state to a circuit, even when the latter is itself re-optimized. The same construction is available for refinement: the n_s slices of Eq. (3) compiled in Sec. V B form a parameterized circuit whose angles are fixed by $(s, \delta t)$, and an *optimized refinement* with a few layers per stage would trade the one asset the adiabatic path has over the UCCSD input in Table II — freedom from classical optimization and training measurements — for a gate count that could fall well below the 4.96×10^4 CNOTs of the $T = 30$ path. Whether a warm-started, physically structured ansatz of this kind escapes the trainability problems of a generic one is the question Ref. [26] raises for lattices; the pairing model, whose optimum is known exactly at every resolution, is the place to settle it, and we list it among the extensions below.

d. Hardware requirements. The Trotterized pipeline quantifies the dominant coherent error of an ideal device, and the requirements it implies are explicit: $2k+1$ qubits (9 at $k=4$, 17 at $k=8$; $c-1$ more for the shared-flag coarse term of Sec. V B); a mid-circuit measurement with reset and reuse of one ancilla; arbitrary-angle R_z rotations; and a coherent budget that ranges from 192 two-qubit gates for the cheapest route of Table II through 10^4 per accepted preparation for the unfiltered HF ramp to 10^5 for the refined input with six cycles at eight qubits, and ten times these at sixteen, on top of the energy scan and the repetitions needed to estimate acceptance probabilities. At the two-qubit error rates of large-scale experiments today — of order 10^{-2} on the superconducting processor of Ref. [53] and $(0.8\text{–}2)\times10^{-3}$ on trapped-ion processors [54, 55] — a 10^4-gate run suffers of order ten faults even on the best of these, so none of these budgets is executable with physical qubits without substantial error mitigation. Early logical-qubit demonstrations on neutral-atom arrays have executed circuits on tens of logical qubits with error detection and mid-circuit readout [56, 57], and the rodeo algorithm has been run in two-qubit form on trapped-ion and superconducting hardware [20, 22]. A proportionate near-term target is a mitigated $k=3$ demonstration — the $2\to3$ chain on seven qubits or the $3\to4$ chain on nine, with a short refinement schedule and one or two rodeo cycles at coarse δt, i.e. 10^3–10^4 two-qubit gates — where the collapse of the acceptance under a corrupted evolution doubles as a hardware diagnostic; the full $k=4$ budget is a workload for the first generation of logical-qubit machines such as the neutral-atom system planned by the QuNorth initiative [58], and Table II is offered as its specification.

e. Concluding remarks and perspectives. The present results establish, on a fully controlled example, how resolution refinement followed by the rodeo algorithm behaves at the circuit level — every stage compiled, counted and validated against exact results — and what it is worth: it prepares the correlated ground state to the accuracy of the Trotterized evolution without any variational optimization, at a gate cost that, for the pairing model, matches that of an adiabatic ramp from the same determinant to within the embedded coarse term at every size studied, exceeds that of a fixed variational input at eight but not at sixteen qubits, and

exceeds that of an adaptively grown variational input by two orders of magnitude at both. Three extensions would sharpen the assessment. First, comparators and compilation: a qubit-ADAPT run with a Pauli-string pool, a pair-encoded (one qubit per level) variational and filtering baseline and a symmetry-projected BCS-like input, an optimized refinement in the sense of Ref. [26] — the Trotterized path of Eq. (3) as a warm-started variational circuit with a few layers per stage — and a compilation effort applied equally to every route (merged half-steps, optimized excitation circuits, the shared-flag coarse term, routing on a concrete connectivity), the search of Table V being then extended to the filter width and step and to the shot budgets of training and energy search amortized over the number of states requested. Second, Hamiltonians for which single-reference variational inputs fail — shell-model interactions with competing deformations, or the strong-coupling regime of Sec. IV D treated with a nontrivial coarse space — which is where the trainability argument for refinement would be tested rather than asserted; the first step in this direction is a seniority-breaking particle-hole interaction, which degrades the single-reference ansatz with the strength of the pair breaking and forces the full $2k$-qubit register. Third, hardware: a mitigated seven- or nine-qubit demonstration on present devices, where the acceptance statistics of the filter double as a diagnostic of the compiled evolution, followed by the full $k = 4$ budget on an early logical-qubit machine. The pairing model, exactly solvable at every resolution, remains the calibration standard for all three.

CODE AVAILABILITY

The complete implementation — pairing Hamiltonian, gate-level UCCSD–VQE, prolongation, adiabatic refinement, the ancilla-based rodeo algorithm, and the seniority-zero pair-basis implementation used for the sixteen-qubit study of Sec. V E (validated against the full-sector code) — is provided as a tested Python package and a single Jupyter notebook that reproduces every figure and number in this article and in its Supplemental Material [59] [60].

ACKNOWLEDGMENTS

The work of C.G. and G.F.L. acknowledge funding from the European Union's Horizon Europe research and innovation program under the Marie Skłodowska-Curie Grant Agreement No. 101126636. M.H.-J. is partially supported by U.S. National Science Foundation (NSF) Grants PHY-1404159 and PHY-2310020. The authors acknowledge Anthropic's Claude for assistance with literature search and coding.

[1] D. Beck *et al.*, *Quantum information science and technology for nuclear physics. Input into U.S. long-range planning, 2023*, arXiv:2303.00113 (2023).
[2] C. W. Bauer, Z. Davoudi, N. Klco, M. J. Savage, *Quantum simulation of fundamental particles and forces*, Nat. Rev. Phys. **5**, 420 (2023).
[3] D. Lee, *Quantum techniques for eigenvalue problems*, Eur. Phys. J. A **59**, 275 (2023).
[4] E. Ovrum, M. Hjorth-Jensen, *Quantum computation algorithm for many-body studies*, arXiv:0705.1928 (2007).
[5] M. Q. Hlatshwayo, Y. Zhang, H. Wibowo, R. LaRose, D. Lacroix, E. Litvinova, *Simulating excited states of the Lipkin model on a quantum computer*, Phys. Rev. C **106**, 024319 (2022).
[6] I. Stetcu, A. Baroni, J. Carlson, *Variational approaches to constructing the many-body nuclear ground state for quantum computing*, Phys. Rev. C **105**, 064308 (2022).
[7] A. Pérez-Obiol, A. M. Romero, J. Menéndez, A. Rios, A. García-Sáez, B. Juliá-Díaz, *Nuclear shell-model simulation in digital quantum computers*, Sci. Rep. **13**, 12291 (2023).
[8] C. Jiang, J. Pei, *Quantum computing of the pairing Hamiltonian at finite temperature*, Phys. Rev. C **107**, 044308 (2023).
[9] J. Zhang, D. Lacroix, Y. Beaujeault-Taudière, *Neutron-proton pairing correlations described on quantum computers*, Phys. Rev. C **110**, 064320 (2024).
[10] E. A. Ruiz Guzman, *Symmetry breaking and restoration for many-body problems treated on quantum computers*, PhD thesis, Université Paris-Saclay (2023); arXiv:2310.17996.
[11] A. Peruzzo, J. McClean, P. Shadbolt, M.-H. Yung, X.-Q. Zhou, P. J. Love, A. Aspuru-Guzik, J. L. O'Brien, *A variational eigenvalue solver on a photonic quantum processor*, Nat. Commun. **5**, 4213 (2014).
[12] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, H. Neven, *Barren plateaus in quantum neural network training landscapes*, Nat. Commun. **9**, 4812 (2018).
[13] M. Larocca, S. Thanasilp, S. Wang, K. Sharma, J. Biamonte, P. J. Coles, L. Cincio, J. R. McClean, Z. Holmes, M. Cerezo, *Barren plateaus in variational quantum computing*, Nat. Rev. Phys. **7**, 174 (2025).
[14] T. Albash, D. A. Lidar, *Adiabatic quantum computation*, Rev. Mod. Phys. **90**, 015002 (2018).
[15] D. S. Abrams, S. Lloyd, *Quantum algorithm providing exponential speed increase for finding eigenvalues and eigenvectors*, Phys. Rev. Lett. **83**, 5162 (1999).
[16] L. Lin, Y. Tong, *Near-optimal ground state preparation*, Quantum **4**, 372 (2020).
[17] Y. Dong, L. Lin, Y. Tong, *Ground-state preparation and energy estimation on early fault-tolerant quantum computers via quantum eigenvalue transformation of unitary matrices*, PRX Quantum **3**, 040305 (2022).
[18] I. Stetcu, A. Baroni, J. Carlson, *Projection algorithm for state preparation on quantum computers*, Phys. Rev. C

108, L031306 (2023).
[19] K. Choi, D. Lee, J. Bonitati, Z. Qian, J. Watkins, *Rodeo algorithm for quantum computing*, Phys. Rev. Lett. **127**, 040505 (2021).
[20] Z. Qian, J. Watkins, G. Given, J. Bonitati, K. Choi, D. Lee, *Demonstration of the rodeo algorithm on a quantum computer*, Eur. Phys. J. A **60**, 151 (2024).
[21] T. D. Cohen, H. Oh, *Optimizing the rodeo projection algorithm*, Phys. Rev. A **108**, 032422 (2023).
[22] M. Bee-Lindgren, Z. Qian, M. DeCross, N. C. Brown, C. N. Gilbreth, J. Watkins, X. Zhang, D. Lee, *Controlled gate networks: theory and application to eigenvalue estimation*, Eur. Phys. J. A **61**, 263 (2025); arXiv:2208.13557.
[23] M. Patkowski, O. Ayyildiz, K. Hunt, N. Jansen, D. Lee, *Improved rodeo algorithm performance for spectral functions and state preparation*, arXiv:2602.05978 (2026).
[24] M. Patkowski, O. Ayyildiz, M. Kebrič, K. L. C. Hunt, D. Lee, *Hierarchical fusion method for scalable quantum eigenstate preparation*, Phys. Rev. A **113**, 052442 (2026); arXiv:2510.19039.
[25] S. Bogner, H. Hergert, M. Hjorth-Jensen, R. LaRose, D. Lee, M. Patkowski, *Quantum state preparation with resolution refinement*, Phys. Lett. B **875**, 140363 (2026); arXiv:2511.14732.
[26] J. P. Sadia, M. Hjorth-Jensen, D. Lee, R. LaRose, *Digital quantum state preparation by nucleation*, arXiv:2609.37527 (2026).
[27] D. Lacroix, *Symmetry assisted preparation of entangled many-body states on a quantum computer*, Phys. Rev. Lett. **125**, 230502 (2020).
[28] E. A. Ruiz Guzman, D. Lacroix, *Restoring broken symmetries using quantum search "oracles"*, Phys. Rev. C **107**, 034310 (2023).
[29] P. Siwach, D. Lacroix, *Filtering states with total spin on a quantum computer*, Phys. Rev. A **104**, 062435 (2021).
[30] E. A. Ruiz Guzman, D. Lacroix, *Restoring symmetries in quantum computing using Classical Shadows*, Eur. Phys. J. A **60**, 112 (2024).
[31] E. A. Ruiz Guzman, D. Lacroix, *Accessing ground-state and excited-state energies in a many-body system after symmetry restoration using quantum computers*, Phys. Rev. C **105**, 024324 (2022).
[32] E. A. Ruiz Guzman, D. Lacroix, *Calculation of generating function in many-body systems with quantum computers: technical challenges and use in hybrid quantum-classical methods*, arXiv:2104.08181 (2021).
[33] D. Lacroix, E. A. Ruiz Guzman, P. Siwach, *Symmetry breaking/symmetry preserving circuits and symmetry restoration on quantum computers: A quantum many-body perspective*, Eur. Phys. J. A **59**, 3 (2023).
[34] T. Ayral, P. Besserve, D. Lacroix, E. A. Ruiz Guzman, *Quantum computing with and for many-body physics*, Eur. Phys. J. A **59**, 227 (2023).
[35] J. Zhang, D. Lacroix, *Excited states from ADAPT-VQE convergence path in many-body problems: application to nuclear pairing problem and H_4 molecule dissociation*, Phys. Lett. B **869**, 139841 (2025).
[36] Y. Beaujeault-Taudière, D. Lacroix, *Solving the Lipkin model using quantum computers with two qubits only with a hybrid quantum-classical technique based on the generator coordinate method*, Phys. Rev. C **109**, 024327 (2024).
[37] S. Aychet-Claisse, D. Lacroix, V. Somà, J. Zhang, *Quantum simulations of Green's functions for small superfluid systems*, Phys. Rev. C **113**, 044324 (2026); arXiv:2509.02272.
[38] M. Hjorth-Jensen, M. P. Lombardo, U. van Kolck (eds.), *An Advanced Course in Computational Nuclear Physics*, Lecture Notes in Physics **936**, Springer (2017).
[39] R. W. Richardson, *A restricted class of exact eigenstates of the pairing-force Hamiltonian*, Phys. Lett. **3**, 277 (1963).
[40] R. W. Richardson, N. Sherman, *Exact eigenstates of the pairing-force Hamiltonian*, Nucl. Phys. **52**, 221 (1964).
[41] J. Dukelsky, S. Pittel, G. Sierra, *Colloquium: Exactly solvable Richardson-Gaudin models for many-body quantum systems*, Rev. Mod. Phys. **76**, 643 (2004).
[42] J. Lee, W. J. Huggins, M. Head-Gordon, K. B. Whaley, *Generalized unitary coupled cluster wavefunctions for quantum computation*, J. Chem. Theory Comput. **15**, 311 (2019).
[43] J. Tilly, H. Chen, S. Cao, D. Picozzi, K. Setia, Y. Li, E. Grant, L. Wossnig, I. Rungger, G. H. Booth, J. Tennyson, *The variational quantum eigensolver: a review of methods and best practices*, Phys. Rep. **986**, 1 (2022).
[44] S. Jansen, M.-B. Ruskai, R. Seiler, *Bounds for the adiabatic approximation with applications to quantum computation*, J. Math. Phys. **48**, 102111 (2007).
[45] L. K. Kovalsky, F. A. Calderon-Vargas, M. D. Grace, A. B. Magann, J. B. Larsen, A. D. Baczewski, M. Sarovar, *Self-healing of Trotter error in digital adiabatic state preparation*, Phys. Rev. Lett. **131**, 060602 (2023).
[46] H. F. Trotter, *On the product of semi-groups of operators*, Proc. Am. Math. Soc. **10**, 545 (1959).
[47] M. Suzuki, *Generalized Trotter's formula and systematic approximants of exponential operators*, Commun. Math. Phys. **51**, 183 (1976).
[48] A. M. Childs, Y. Su, M. C. Tran, N. Wiebe, S. Zhu, *Theory of Trotter error with commutator scaling*, Phys. Rev. X **11**, 011020 (2021).
[49] A. Kandala, A. Mezzacapo, K. Temme, M. Takita, M. Brink, J. M. Chow, J. M. Gambetta, *Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets*, Nature **549**, 242 (2017).
[50] M. A. Nielsen, I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press (2010).
[51] H. R. Grimsley, S. E. Economou, E. Barnes, N. J. Mayhall, *An adaptive variational algorithm for exact molecular simulations on a quantum computer*, Nat. Commun. **10**, 3007 (2019).
[52] H. L. Tang, V. O. Shkolnikov, G. S. Barron, H. R. Grimsley, N. J. Mayhall, E. Barnes, S. E. Economou, *Qubit-ADAPT-VQE: an adaptive algorithm for constructing hardware-efficient ansätze on a quantum processor*, PRX Quantum **2**, 020310 (2021).
[53] Y. Kim, A. Eddins, S. Anand, K. X. Wei, E. van den Berg, S. Rosenblatt, H. Nayfeh, Y. Wu, M. Zaletel, K. Temme, A. Kandala, *Evidence for the utility of quantum computing before fault tolerance*, Nature **618**, 500 (2023).
[54] S. A. Moses *et al.*, *A race-track trapped-ion quantum processor*, Phys. Rev. X **13**, 041052 (2023).
[55] A. Ransford *et al.*, *Helios: a 98-qubit trapped-ion quantum computer*, arXiv:2511.05465 (2025).
[56] D. Bluvstein *et al.*, *Logical quantum processor based on reconfigurable atom arrays*, Nature **626**, 58 (2024).

[57] B. W. Reichardt *et al.*, *Logical computation demonstrated with a neutral atom quantum processor*, arXiv:2411.11822 (2024).

[58] QuNorth, *Magne: a level-2 neutral-atom quantum computer*, `https://qunorth.com` (accessed July 2026).

[59] See Supplemental Material at [URL will be inserted by publisher] for the coarse-solver comparison at larger basis size, the refinement energies and the $3 \to 4$ chain, the shift-scan fits and the response norm along the path, the evaluation of the sufficient adiabatic-time and Trotter-step conditions, the rodeo energy scan and its statistics, the operating points behind the quoted fidelities, the other model-space sizes, particle numbers and observables, and the validation of the pair-basis implementation used at sixteen qubits.

[60] Repository `pairing-eigenq`, [URL to be inserted]; archived version: Zenodo, [DOI to be inserted].